

Superdiffusion and nested oscillations in visual search dynamics

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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

Jude Luca Metcalf

October 22, 2026

Abstract

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Introduction

The brain is among the most complex systems in science, and one of its most important features is its ability to build a rich understanding of the world from a small, selectively sampled fraction of the available input. Nowhere is this clearer than in vision. In humans, an estimated one in five cortical neurons is devoted to visual processing and control [1, 2], yet despite this computational power, the efficiency of information processed is extreme: for every 1 MB of information reported by the retina, only about 40 bit are consciously processed [3–5]. The visual system must therefore be highly selective, and it achieves this by acting rather than passively receiving. High-acuity vision is confined to the fovea, so to build a useful representation of the visible world, an observer must move their eyes, directing the fovea towards successive locations of interest. These rapid ballistic movements, called saccades, organise periods of relative stillness known as fixations during which the brain gathers visual information, and together they form a sequence of decisions about where to look next, made several times every second during searching [6, 7]. Each decision depends on what has already been seen, what is currently in view, and the goals of the observer. Eye movements are thus the external output of a closed perception–action loop in which the brain selects the very data on which it will subsequently make decisions about further perception. This process of decision-making informing further perception is called active vision.

This makes gaze an incredibly valuable observable for understanding the brain. Gaze trajectories provide a direct, low-dimensional, continuous, and non-invasive proxy for neural attention, so the spatiotemporal dynamics of attention become direct to measure [8–16]. The dynamics of visual attention is driven by many neural processes that are otherwise difficult to access experimentally: the evaluation of stimulus salience [10, 17–19], working memory through inhibition of return (IoR) to explored visual regions [20–24], as well as more generally in complex tasks integrating top-down goals [25–27], and the intrinsic timescales of neural activity through regions of visual processing and control (importantly the V1 and the superior colliculus) [8, 9, 28, 29]. Even the low-amplitude movements made during fixation, once dismissed as noise, carry structure reflecting the underlying oculomotor and neural machinery [30–35], and are valuable in gathering information during fixation [36–43].

Visual search, in which an observer scans a scene for a target or simply explores it with a determined goal, is among the most demanding and informative settings for studying decision-making in the brain. The subject must balance an exploration–exploitation trade-off, deciding when to remain sampling promising regions and when to discover new ones, while avoiding wasteful re-exploration of depleted regions. This must be done efficiently and robustly within the brain, as the cost of neural computation is high and the brain must be trained to perform well to many tasks [44–48]. The resulting trajectories which are found to be most efficient are rich and structured, with characteristic spatial statistics, temporal correlations, and preferred timescales [8, 9, 11, 40, 49–51]. Understanding the mechanisms behind them offers a window not only onto vision, but onto the general strategies the brain uses to explore, sample, and decide under uncertainty [52–58], such as has been developed under the theory of active inference [59–61]. Such efficient and structured dynamics also outline the importance of deficiencies in eye-movement patterns associated with a range of neurological and psychiatric conditions [62–69]. This finding imbues these measurements with a broad value in science, towards informing biomarkers of global brain function that aggregate many regions involved in visual search.

The rich structure of gaze trajectories now invites a suitably complex physical description. Gaze can be modelled as a patterned stochastic process which has been aggregated from a complex adaptive system in the network of neural activity in a critical regime with memory effects [11, 57]. Interpreting these dynamics within the informative visual landscape optimising a target search is the driving motivation for this thesis. This model makes it possible to ask quantitative questions about gaze dynamics: how memory effects and relative adaptive timescales improve dynamical sampling, the structure of the heavy-tailed statistics of noise-driven motion, and what enforces characteristic sampling timescales. Methods from statistical physics, nonlinear dynamical systems, and non-equilibrium thermodynamics

have already provided new ways to characterise and explain the collective and dynamical properties of neural computation [CITE], and they offer a language for computational neursoscience that is both explanatory and precise; such a well-defined mathematical model then permits an understanding of the system which drives these dynamics, the attention mechanism of the brain itself. Gaze is also a rare example of a driven, history-dependent stochastic system for which high-resolution data are readily available, making it a natural testbed for such theories.

Existing work has approached gaze dynamics largely one scale at a time, both in terms of space and time. Studies of large-scale saccades characterise their timing, amplitudes, and kinematics [6, 20, 35]. At the scale of whole search trajectories, stochastic models describe global strategies with heavy-tailed, Lévy-like exploration [70–72], or else through global information-maximising strategies [18, 19, 73–80]. The faster- and shorter spatiotemporal scales of fixational eye movements are then treated as a separate subject, through stochastic descriptions of their high-frequency noise statistics [7, 81, 82] or through their perceptual role in rapid information-gathering [33, 83], as well as the low-amplitude drifts during fixation [30, 34, 35, 83]. The various timescales of attention more broadly, such as theta rhythmicity, form a further literature rooted in neural recordings and behaviour [8, 9, 28]. Each account is valuable on its own terms, but they are developed with different formalisms, different data, and often different assumptions about what the eye is doing.

Alongside these, the dominant recent approach to modelling human gaze comes from machine learning. A large body of work trains deep networks to predict either gaze trajectories [73, 77, 84–87] or saliency maps indicating where observers are likely to look [17–19, 79, 88, 89]. Most notably, DeepGaze III [75] predicts human fixation locations with high accuracy given the image, the fixation history, and systematic biases in how people search [75, 88, 90]. Such models have been used to test hypotheses about search, showing for example that simple low-level contrast is often sufficient to predict fixations [76], and that late fixations become dominated by intrinsic gaze biases and decouple from image content [75]. Yet these networks are optimised to predict where the next fixation will fall, not why gaze behaves as it does. Their internal representations are difficult to interpret, they do not readily connect to biological mechanisms, and they say little about the temporal organisation of search as a dynamical process [CITE].

Each of these approaches leaves a gap; both between biological realistic accounts and empirical accuracy, and between the fine and coarse scales of perception which should each be captured under a single theory. Saccades and fixations are not independent processes: the regularity, size, and duration of saccades are influenced strongly by information gathered during fixation [CITE]. Models built for a single scale cannot capture this coupling, and their parameters are fitted separately and carry no common physical interpretation. As such, one cannot simply combine such theories into a patchwork theory, as this will not accurately describe the neural control of gaze dynamics. The combined large- and small-scale movements are also rarely measured together, so there are few quantitative constraints on how a unified model should behave across scales [49]. As a result, we lack a single, biologically motivated model in which the heavy-tailed spatial statistics of search, the rhythmic timing of saccades, memory of explored regions, and the structured dynamics of fixation arise from the same underlying biologically-informed mechanism. As such, a unified mechanistic account of search dynamics, going beyond fixation prediction, remains largely unexplored.

Such an account must reconcile the phenomena above while remaining biologically plausible, parsimonious, and capable of generating testable predictions about novel or unexplained aspects of visual search. It should treat these features as different aspects of one stochastic, history-dependent process rather than as separate phenomena patched together. This thesis aims to unify these behaviours within a single functional, dynamical model of human visual search, linking the statistics of gaze to the principles by which the brain actively samples its environment.

Chapter 1: Survey of the Literature

In this thesis, we aim to unify the rich gaze dynamics found during active visual search under a single physics-inspired model, admitting a deeper understanding of neural computation. There is growing success in turning methods from statistical physics and nonlinear dynamics towards the area of computational neuroscience. In particular, the rich scale-free activity of neural circuits inspires similar analysis to unify the dynamics of gaze trajectories across spatiotemporal scales, and present a physics-inspired dynamical understanding of neural computation in search tasks.

In Section 1.1, we review the background for developing a scientific understanding of the dynamics of visual gaze; the data we present in this thesis, the unique spatiotemporal patterns of such data, and the previous methods used to model complex dynamics found in gaze trajectories are each described. In Section 1.2, we introduce the theoretical framework of stochastic modelling, Lévy motion, and the anomalous superdiffusion found elsewhere in complex dynamical systems. Section 1.3 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts; providing the key motivation for deriving a complete mathematical understanding of a gaze trajectory as a proxy for dynamical attention in the brain.

1.1 Behavioural data

To ground the physical background of this thesis, the present section reviews the physical system we aim to model; the eye-tracking signal which characterises human visual attention. During active vision, the visual system does not operate as a passive receiver of sensory input, but rather as an active, dynamic sampler of a changing environment [59, 91]. This active sensing manifests across two coupled physical observables: the internal dynamics of rich neural oscillatory networks, and the external, mechanical execution of gaze trajectories. Crucially, these domains inform and drive each other, and each exhibit complex, scale-free dynamics in their activity. Characterised by background spectra with power law tails and anomalous diffusion, the visual system is capable of devoting attention to many spatiotemporal scales of event. Viewing these biological phenomena through the lens of statistical mechanics provides a rigorous foundation for mathematically modelling the visual system. This section establishes these empirical constraints by first examining the functional role of neural frequency bands and theta-gamma coupling (Section 1.1.3), before detailing the multi-scale, stochastic regimes of human gaze trajectories (Section 1.1.1). Together, these behavioural patterns motivate the fractional dynamics given by the FNS equations of motion, and the developments upon that model in the subsequent sections.

1.1.1 Gaze trajectories

Healthy human gaze trajectories are well characterised by a small set of dynamical patterns. The most studied pattern of gaze trajectories are their division into periods of fixation, interspersed by near-instantaneous jumps between fixations, these jumps being known as saccades [6, 69, 72, 92]. Importantly, this behaviour of local movements interspersed with large discontinuous jumps suggest modelling eye movements through a Lévy flight. This paradigm has a limited history in the literature, and fitting data to a simple phenomenological Lévy flight parameterised by the stable parameter α has found some limited use as a biomarker for ADHD [62]. Broader saccade metrics have been found to be valuable as non-invasive biomarkers for broader neural disorders, including schizophrenia [63–65, 67], autism [68, 93], OCD [68], bipolar disorder [68, 93], and Parkinson's disease [68, 93]. The dynamics of fixation have received less scholarly attention as a biomarker of cognition. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [7, 22, 31, 32]. Detailed examinations and systematic reviews of these behaviours have not considered their computational role in active sensing and efficient search. The work presented in this research strongly motivate a closer and more biologically informed understanding of the dynamics of gaze trajectories, across a number of spatial and temporal scales, to diagnose and understand the function of global functioning of the brain, much of which controls the visual system.

1.1.2 Existing models of visual searching dynamics

Progress towards modelling gaze dynamics during visual search spans physics, neuroscience, and machine learning. As such, existing models have been built with very different aims and methods. Cognitive and biologically inspired models explain where attention is drawn through biologically-inspired saliency maps and top-down guidance [94–96]. Normative models ask what an optimally informative searcher would do, selecting saccades by Bayesian belief updating and entropy reduction [97]. Physics-inspired models may describe gaze through a dynamical system, through variational least-action principles [70] or through stochastic descriptions of fixational drift [7]. More recently, deep networks trained on eye-tracking data predict human scanpaths on natural images with high accuracy [75, 85].

Each approach captures an important aspect of search, but so-far models must focus on a specific feature and scale of gaze dynamics. Most treat search as a sequence of discrete saccades and neglect continuous-time dynamics; fixational activity is modelled separately from saccades, and the most accurate predictors are not designed to be biologically interpretable. No model among those reviewed combines continuous-time dynamics, saccadic jumps, fixational motion, and an explicit informational account of search. The present model draws on three of these approaches: the information kernel of the ideal Bayesian searcher, the second-order saliency-driven dynamics of the least-action model, and the structured fixational motion of the self-avoiding walk. A full comparison of the most influential and successful models, along with a comparison of their core domains, aims, and key features, is given in Table A.1 in Appendix A.

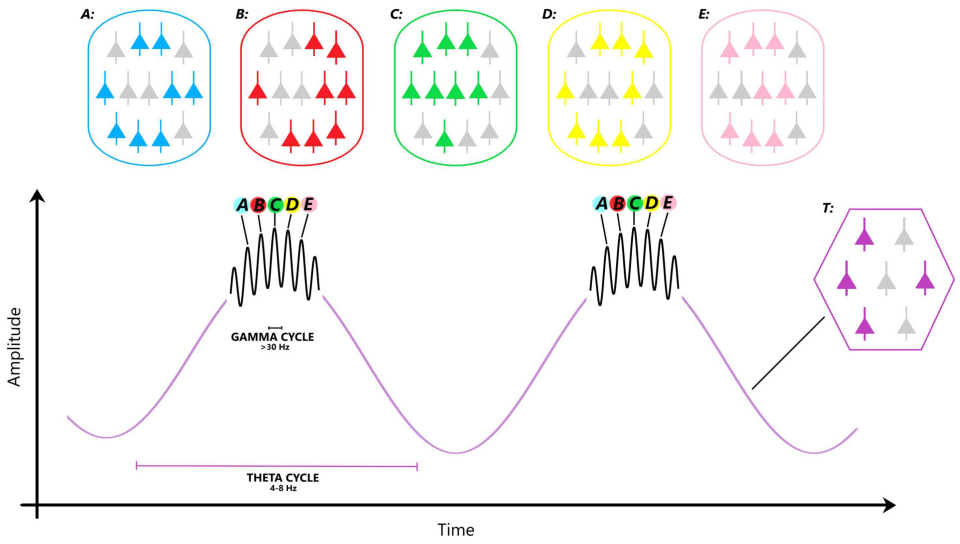


Figure 1.1: A schematic representation of theta-gamma coupling (TGC). Here, neural population patterns $\mathbf{A} - \mathbf{E}$ are activated in sequence on the timescale of a gamma oscillation $f > 30$Hz, triggered by the phase of a modulating theta wave of neuron population $\mathbf{T}$ on the theta timescale of $4 - 8$Hz. Such a formulation of TGC as correctly ordering signals has its foundations in the work of Lisman and Jensen [98], and was subsequently expanded to describe the dynamic neural mechanism during sensory exploration to modulate close attention upon individual centres of focus [99]. *Schematic adapted from figure 2 of Ursino and Pirazzini [100]*

1.1.3 Timescales of neural computation

Throughout much of this thesis, links are drawn between timescales of interactions and dynamics in the FNS model and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data. Divisions are only approximate, but for this thesis these bands

are: delta (0.1–3 Hz), theta (3–6 Hz), alpha (6–14 Hz), beta (14–30 Hz), and gamma (40–100 Hz) [101, 102]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to neural computation; broadly, theta rhythms have been found to modulate and organise bursts in computational gamma-scale neural activity [9, 57, 103–106]. Theta rhythms have been shown to modulate attention during auditory tasks [14, 107], visual search tasks [8, 9, 108], and when analysing the broad attentional mechanism of the brain [28, 101, 102].

Theta-rhythmic cycles are finding strong interpretation in the cognitive neuroscience literature as organising and communicating information sampled from the environment for integration into the internal model of the brain [9, 103–106]. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [14, 28], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [8, 9]. Accordingly, the firing patterns which define the lowest level of information carried out in the brain are regularly found to be gamma in scale, with broadband signals being found within 40–100 Hz in a wide range of measurements of neural activity [100, 109–112]. The coupling between the phase of theta activity and the amplitude of gamma activity is known as theta-gamma coupling (TGC). This phenomenon, a specific form of phase-amplitude coupling (PAC), is schematically represented in Figure 1.1. The evidence and theory behind TGC are described in a systematic review by Ursino and Pirazzini [100]. This thesis provides a computation view of the function of TGC in the within the search dynamics of gaze as modulating periods of fixational attention during which information is dynamically sampled.

1.2 Diffusive processes

In formalising empirical observations of human gaze and neural activity into a mathematical model, this thesis relies on the framework of non-equilibrium statistical mechanics and stochastic calculus. When a system is driven by a massive number of microscopic interactions—such as the complex interconnected firing of a neural population—modelling the exact deterministic dynamics becomes intractable. Instead, statistical physics provides the tools to coarse-grain these high-dimensional systems into low-dimensional stochastic differential equations. This section establishes the mathematical progression necessary to model scale-free visual inference. We first review the classical paradigm of normal diffusion and Brownian motion (**??**), which serves as the foundational model for standard diffusive physical systems. We then examine how systems of interest frequently violate the classical assumptions of normal diffusion and Brownian motion, leading to an understanding of the presence and role of anomalous diffusion in many complex dynamical systems (Section 1.2.2). Finally, we introduce Lévy stable motion (Section 1.2.3), a class of superdiffusive fractional stochastic processes that mathematically capture the heavy-tailed, intermittent dynamics characteristic of optimal visual search and neural activity.

1.2.1 Brownian motion and normal diffusion

Statistical modelling has been fundamental in understanding complex dynamical systems: separating the dynamics of macroscopic observables into deterministic terms and stochastic terms reduces chaotic high-dimensional systems to a small number of variables of interest [113, 114]. Stochastic processes are often the most natural, tractable, and accurate modelling tools when examining high-dimensional chaotic systems across physics, economics, geology, climate science, machine learning, and neuroscience [[**CITE**]]. The universality of these stochastic models is understood to be an implication of the central limit theorem: the distribution of the summed effect of a large number of identical and independently distributed (i.i.d) random variables of finite mean and variance will approach a Gaussian [115–117]. The technique of treating a large number of independent coupled deterministic variables as a single stochastic variable turns out to be generalisable to a wide class of problems involving coupled, non-stationary, or diverging-variance processes which are out of equilibrium, such as in the case of neuronal spiking patterns [118, 119]. The most ubiquitous stochastic process is central to statistical physics: the Brownian motion W_t, equivalently called the Wiener process, which consists of stationary and independent Gaussian increments across all non-overlapping time intervals. The Brownian motion is

a very simple and analytically robust model of the coarse-grained effect of a high dimensional system of contributions which are flattened to a small number of observable dimensions. A Brownian motion is defined only by its timescale, with a variance over trajectories after time t of t itself. Realisations of Brownian motion yield time series' which are almost surely continuous but nowhere differentiable. Classical examples of processes modelled by the Brownian motion are: the movement of a macroscopic particle in a non-turbulent fluid [113], the gravitational influence of an orbiting collection of stars on supermassive black holes [120, 121], and (arguably) the movement of low-volatility stock prices [114]. The generalisation of the Brownian motion for heavy-tailed i.i.d variables is a Lévy flight, and is described in detail in Section 1.2.3, and Figure 1.2 compares the two processes.

To include random variables such as Brownian motion in dynamical systems, physicists and mathematicians have developed the tools of stochastic calculus. Brownian motion of a variable $\mathbf{x}_t$ through time t about a potential landscape $V(\mathbf{x})$ can be described in terms of the first order Langevin equation:

$$d\mathbf{x}_t = -\gamma \nabla V(\mathbf{x}_t)dt + \sqrt{2\gamma}dW_t \tag{1.1}$$

where the constant γ describes the noise strength of the Brownian motion [122–125] and dW_t is a Brownian motion. Although the dynamics of $\mathbf{x}_t$ is described stochastically, the dynamics of the corresponding probability distribution $P(\mathbf{x}, t)$ is deterministic and can be described exactly. Such an equation of motion of $P(\mathbf{x}, t)$ under any stochastic dynamics defining $\mathbf{x}_t$ is called the Fokker-Planck equation [126–131]. In the case of Equation (1.1), the Fokker-Planck equation is [126, 127, 130]

$$\partial_t P(\mathbf{x}, t) = \frac{\gamma}{2} \nabla \cdot (P(\mathbf{x}, t)\nabla V(\mathbf{x}) + \nabla P(\mathbf{x}, t)) \tag{1.2}$$

whose unique stationary distribution is the Boltzmann-Gibbs distribution

$$\pi(\mathbf{x}) \propto \exp[-V(\mathbf{x})] \tag{1.3}$$

which yields the important relationship, where a Langevin potential can be constructed from a normalised target density via $V(\mathbf{x}) = -\ln \pi(\mathbf{x})$, which we will make use of in this thesis in constructing saliency maps from empirical data where only the likelihood of fixations $\pi(\mathbf{x})$) is given. Importantly, the Langevin system Fokker-Planck equation 1.2 will describe the diffusion of a sharply peaked initial distribution as it widens, and the statistics of this diffusion from a sharp initial condition define normal diffusion as follows. The solution to the Fokker-Planck equation from an exact initial condition $P(\mathbf{x}, 0) = \delta^n(\mathbf{x} - \mathbf{x}_0)$ is dominated over short times by the ∇P term as $\nabla P \gg P\nabla V$, reducing the Fokker-Planck equation to the heat equation $\partial_t P = \frac{\gamma}{2}\nabla^2 P$. As such, the leading-order short-time solution is that of such a heat equation

$$P(\mathbf{x}, \Delta t) \propto \exp\left[-\frac{\|\mathbf{x} - \mathbf{x}_0\|^2}{2(\gamma \Delta t)}\right] \implies \mathrm{Var}[\mathbf{x}(\Delta t)] = \gamma \Delta t \tag{1.4}$$

which demonstrates the interesting behaviour which defines normal diffusion: the short-time scaling of variance in position will be linear, as $\mathrm{Var}[\mathbf{x}_{\Delta t}] = \gamma \Delta t$. Thus γ is often called the timescale of the Brownian motion, as it defines the rate of linear diffusion of variance with time. This linear scaling of the variance of trajectories with time defines normal diffusion [132–135], and understanding those systems whose random motion does not reproduce such normal diffusion is crucial in understanding complex dynamical diffusion.

1.2.2 Anomalous diffusion

The universality of normal diffusion in systems which are driven by Brownian motion as above has been studied closely, most influentially as in the movement of particles in a fluid [121, 132, 136, 137]. These dynamics inform the definition of normal diffusion through this variance scaling with time, rigorously calculated as the mean squared displacement (MSD) $\langle \|\Delta \mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t$. To study the universality of such normally diffusive processes, the continuous time random walk (CTRW) model was developed, in which a trajectory has subsequent step sizes and durations drawn from finite variance distributions.

Such models will universally reproduce this normal diffusive scaling property of $\langle \|\Delta \mathbf{x}(t)\| \rangle \propto t$, and as a result of the central limit theorem, will approach Brownian motion. However, increasingly interesting physical systems have been found which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle \|\Delta \mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t^{\lambda_2}, \quad \lambda_2 \neq 1 \tag{1.5}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \lambda_2 < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \lambda_2 < 2$ where diffusion scales faster than normal. The case of $\lambda_2 = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$. An important phenomenon often found in diffusive processes is fractal self-similarity. A Brownian motion is self-similar across all scales, and is nowhere smooth, and in physical systems demonstrating anomalous diffusion, self-similarity resulting from criticality is often reported. Later in Section 1.2.3, we will find that it is advantageous in the study of anomalous superdiffusion to capture this multi-scale diffusive self-similarity through an extension of the λ_2 scaling of MSD to a spectrum of absolute moments $\lambda(q)$ which describes the diffusive self-similarity which unifies across many scales of study. A comparison of Brownian motion and Lévy motion which displays their self-similar properties and diffusive scaling through the mean absolute displacement $\langle \|\Delta \mathbf{x}\| \rangle \propto \Delta t^{\lambda_1}$ is shown in Figure 1.2.

Anomalous subdiffusion has been shown to result from heavy-tailed wait-times between interactions [132], such as in trapped fluid flow [138], controlled diffusion through cell membranes [132], and protein mixing [139].

Anomalous superdiffusion

Of great importance in studying the dynamcs of visual search, superdiffusion has been studied in the contexts of complex exploratory biological systems. We model the efficient exploration of gaze dynamics through Lévy superdiffusion, which is the standard dynamical model of superdiffusion in physics and is presented in detail in Section 1.2.3. Such Lévy superdiffusion has a strong link to the problem of designing a strategic search for scarce resources. However the true computational benefit of superdiffusion has become a subject of debate in the literature. On one side, the Lévy foraging hypothesis posits that the standard superdiffusive CTRW, the Lévy walk, is an optimal strategy when searching for sparse and randomly distributed resources. On the other, researchers have proposed that such a model is biologically unrealistic and a result of artificial modelling conditions.

The finding that Lévy superdiffusion is optimal in random search was first reported in a letter to the journal Nature in 1999 by Viswanathan et al. [140]. In the original paper proposing the Lévy foraging hypothesis a Lévy walk with superdiffusion $\langle x(t)^2 \rangle \propto \frac{t^2}{\ln(t)}$ on the transition to ballistic diffusion was found to be optimal in a search for sparse resources [140]. Recently, physicists and biologists have questioned if this result is imposed by an unrealistic modelled task, an argument described most strongly just this year by Pyke, Kepert, and Chivers [141], as well as in general analyses of superdiffusion in [142–144]. However, despite such disagreement, the optimality of the superdiffusive Lévy walk has been reproduced in diverse functional search models [145–147], in a population of diverse searching agents with a range of superdiffusive exponents λ_2 [148] as well as in a large study against several different search conditions [145], and have been reported in the empirical search paths of biological organisms from the flights of drosophila flies [147, 149], albatross' searches [150, 151], and strongly across many marine predators [152, 153].

The result of these analyses yields a complex understanding of Lévy superdiffusion, advantages of Lévy walks have been shown to be model-dependent, with the optimal performance of Lévy walks requiring biologically inaccurate depictions of search paths [141, 144, 154]. These arguments marks a split in the treatment of Lévy walks, with groups studying them as informative baseline functional models of abstract search [140, 145, 148, 154], and others led by Graham Pyke presenting this as a misrepresentation of true intelligent organisms [141, 143, 144]. This thesis provides a biologically realistic and well-motivated superdiffusive search for research into a case for optimal Lévy motion. See Reynolds [142] for an in-depth discussion of a richer view of biological Lévy superdiffusion.

1.2.3 Lévy motion

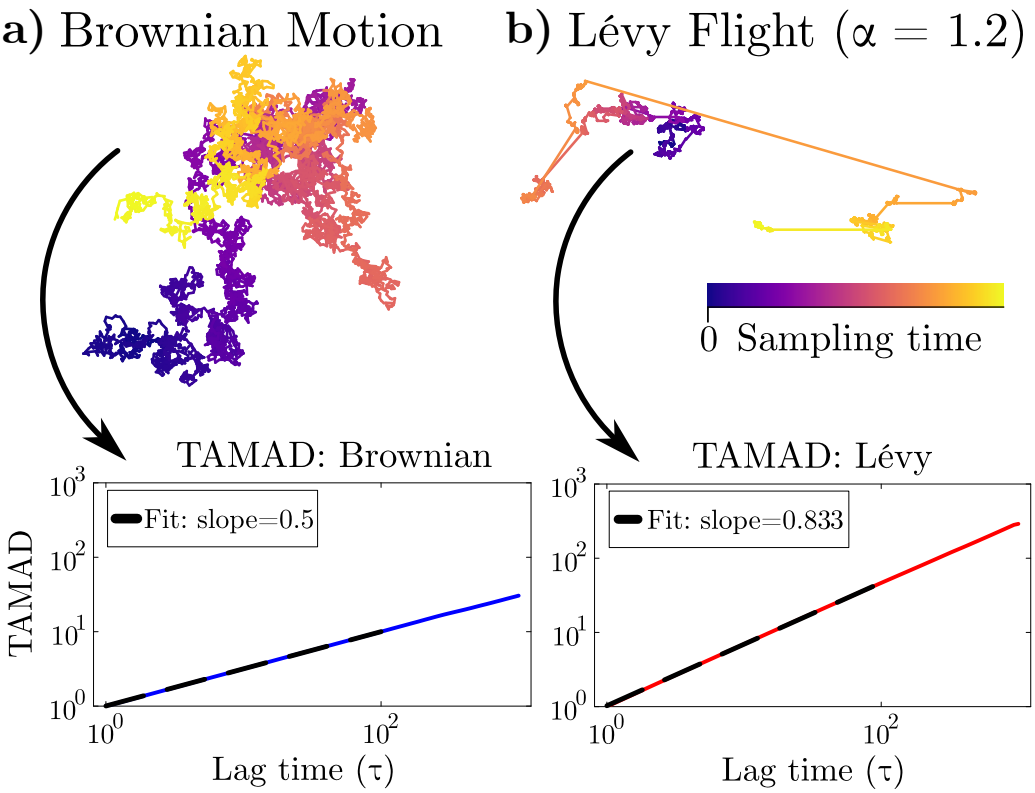


Figure 1.2: Schematic representations of stochastic motion and their respective diffusion. **a)** Schematic representation of a 2D Brownian motion, along with its time-averaged mean absolute displacement (TAMAD), which is the $q = 1$ mean fractional displacement as in Equation (1.7). Normal diffusion is here present as a power law slope of exponent $1/2$, giving diffusive exponent $\beta = 1$ as defined in Equation (1.7). **b)** Schematic of a superdiffusive 2D Lévy flight with characteristic exponent of $\alpha = 1.2$, and a MAD slope of 0.833 corresponding to the theoretical superdiffusive coefficient $\beta = 2/\alpha = 5/3$.

To mathematically capture the intermittent, scale-free nature of visual search, we turn to the Lévy flight: a superdiffusive stochastic process that permits close analysis, due to it's very foundational mathematical definition. Here, the Lévy flight L_t^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric α-stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution), where α is called the characteristic exponent and relates to the exponent of the distributions powerlaw tails. $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables, as a result of the generalised central limit theorem, making them ideal to describe a universality class of superdiffusive processes. The α value defining a particular $\mathcal{S}\alpha\mathcal{S}$ distribution is a result of the asymptotic power-law tail of the density function, with the probability density function following $p(x) \propto |x|^{-1-\alpha}$. As a result, the Lévy flight L_t^α after time t is defined by its fractional scaling properties and alpha-stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.6}$$

where $\overset{d}{=}$ denotes equal in distribution. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaled Brownian Motion $L_t^{\alpha \to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges and has mean 0, whereas if $0 < \alpha \leq 1$, then both the variance and mean diverge. In analysing the distribution of a Lévy flight, one must generalise the standard Langevin Fokker-Planck equation (1.2) to the Lévy motion using fractional calculus to capture the nonlocal transport of the particle. The details of such an analysis are beyond this thesis, however the fractional Fokker-Planck equation is the tool used to understand the superdiffusion, nonlocality, and scale-free behaviour of trajectories of Lévy flights (see the appendices to [57]). Many of the simple analytic properties of the Brownian motion are no longer

true of a Lévy flight. Notably, since these flights have infinite variance in steps sizes, the mean squared displacement diverges. Thus their superdiffusion is better measured using an extension of MSD to fractional absolute displacements, through the definition of the moment spectrum function $\lambda(q)$ as in [133]:

$$\langle \|\Delta \mathbf{x}(t)\|^q \rangle \propto t^{\lambda(q)}, \quad q > 0 \tag{1.7}$$

where q is the order of the fractional absolute displacement, and the power law slope will converge for an α-Lévy flight to $\lambda(q) = q/\alpha$ when $q < \alpha$. As such, for Lévy flights with exponents $\alpha > 1$ superdiffusion is well-characterised by the mean absolute displacement, where $q = 1$ in Equation (1.7).

Importantly, trajectories of a Lévy flight are no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite à gauche) with respect to time, which describes functions which are right continuous with well-defined left limits [155]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [156, 157]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [155]. Importantly, it follows from the Lévy-Itô decomposition theorem that a Lévy flight can be decomposed as the sum of two processes [158]:

1. A compound Poisson process of large jumps with $\|\Delta \mathbf{x}\| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number in any finite time period.

2. A square-integrable martingale (mean 0) jump process with $\|\Delta \mathbf{x}\| < 1$. This process is then finite-variance, and as such is subject to the central limit theorem, meaning that under temporal coarse graining it approaches a Brownian motion.

This division is useful in analysing the process in the context of nonlocal stochastic exploration in Lévy flights, which is modelled by the compound Poisson process of large jumps $|\Delta \mathbf{x}| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta \mathbf{x}| < 1$. In the context of visual searching, such a division naturally inspires the division between large jumps of attention, and periods of prolonged attention. As such, we are interested in why such superdiffusive scale-free dynamics may have a powerful in computational advantage in exploring an abstract unknown landscape.

1.3 Abstract computation

The stochastic dynamical exploration of an information landscape is a powerful paradigm of general computation, and is known as probabilistic computation. Such a landscape in this context is a representation of a target probability distribution. Sampling processes represent and evaluate these distributions through the efficient production of informative stochastic samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods aim to draw sampled independently from a target distribution. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from sparse samples of large multivariate distributions inferred from observational data [55, 159]. This framework has recently gained significant traction as a basic scheme to describe biological computation. The variability in firing patterns of neural populations would thus maintain an internal model of perceptual uncertainty and evolving understanding, giving a functional role to chaotic neural activity [160–162]. This section outlines the paradigm of neural sampling as an interpretation of the variability in neuron firing patterns and visual search gaze trajectories (Section 1.3.1) as well as their extension to describe visual searching as a sampling process (Section 1.3.2).

1.3.1 Neural sampling

The framework of neural sampling posits this paradigm that the variability of cortical neural responses is an important functional feature of biological computation. Otherwise regarded as noise, stochastic neural activity patterns are thus presented as samples drawn from a posterior distribution over possible sensory interpretations [54, 163]. The neural sampling hypothesis has been shown to perform and

predict network-level empirical patterns of neural activity in the V1, and recently by Terada and Toyoizumi in biologically realistic cortical networks, better than alternative theories [54, 56, 163, 164]. Perceptual uncertainty is then encoded by the variability of these cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [54, 55, 163]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of neural computation.

Studies into neural sampling show that variability is not strictly a result of explicit stochastic noise, as traditionally believed. Instead, recent work demonstrates that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models are built to capture [56, 163]. Terada and Toyoizumi link this sampling of a posterior distribution to Bayesian learning through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [56]. Further biologically plausible implementations of such Bayesian sampling are energy-based models (EBMs) which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [58]. The sampling hypothesis has thus given a convincing theory of biological neural computation, with testable and well-supported evidence that neural variability has an important functional role in computation [54, 164]. Traditional Gaussian and Poisson noise models representing these stochastic network dynamics inherently correspond to continuous struggle to capture the full dynamics of complex interconnected inhibitory-excitatory (IE) biological neuronal network [165–167]. Further study into the dynamics of these IE networks has thus motivated further functional models of neural activity.

1.3.2 Visual sampling

This thesis presents visual search tasks as such a paradigm where a sampling model is well motivated, and driven by the underlying neural sampling process as well. The visual system is evolved to actively and efficiently process noisy information and evidence from the external world, and plan future search paths to gather further noisy evidence [17, 52, 97, 168]. In the context of visual searching, visually interesting targets with *saliency* as the equivalent of the target landscape, and where the information is located in a target image. Here, saliency is equivalent to the Boltzmann log-likelihood potential from the probability distribution of fixations $p(\mathbf{x})$

$$S(\mathbf{x}) = -\log(p(\mathbf{x})) \tag{1.8}$$

where visual search paths are most likely to be drawn to is a common technique whic [10]. As such, undirected visual searching of an image will show the areas most salient to the brain, allowing insight into the computational sampling mechanism of the visual system. In particular, search tasks require sophisticated integration of the cost and expected information gain of decisions, which invites fruitful decision analysis [169]. Empirical experiments show that eye movements show good agreement with greedy local information maximising fixations, given well-motivated visual sensitivity based on human physiology [52, 97, 170].

Fractional neural sampling

The FNS dynamics are called fractional, as the non-local dynamics of their distribution require fractional calculus to describe such spatial superdiffusion in Lévy flights [171]. The theory of FNS thus unifies the scale-free nonlocal dynamics of neural firing patterns with the neural sampling framework derived from the ideal Bayesian scheme of active inference. The critical advantage of FNS over other standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—likely a major feature of processed natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [172–175]. Furthermore, the generality of the FNS description permits the application of FNS equations of motion to gaze dynamics, as the motion of the eyes are driven by an IE neural circuit which is expected

to keep key features of the FNS model invariant, such as superdiffusion, heavy-tailed statistics, and nonlocal interactions.

Chapter 2: A Unifying Model

Eye movements during visual search have been thought to be a patchwork of distinct behaviours: nonlocal jumps (saccades), periods of fixation, and fine motion within fixations. This chapter presents a biologically motivated model which demonstrates that these behaviours can arise from a single physical process, and to motivate each dynamical component from first principles rather than treating each scale separately. Uniquely, the presented model captures and predicts the rich nested oscillations and superdiffusion in visual searching dynamics; moreover these dynamical signatures of attentional timescales, gamma-frequency (40–100 Hz) sampling, and superdiffusion take origin from network-level neural activity patterns which have been found in the brain [110]. Inspired by such neural dynamics, we model visual search trajectories as a stochastic process driven by travelling activity waves across the surface of the brain, using the theory of fractional neural sampling (FNS) [11, 57].

The chapter is organised as follows. Section 2.1 defines the model based from FNS dynamics before presenting the search task used as the paradigm throughout this work. Section 2.2 introduces the dynamical-informative kernel which is novel to this work, giving its modelling motivations (Section 2.2.1) and the regular information-driven temporal structure of transitions between regions of interest as they are explored through informational kernel deposition (Section 2.2.3). Section 2.3 derives and quantifies the prediction from neural dynamics that localised trajectories produce high-frequency simple harmonic oscillations. Section 2.4 treats the defining characteristic of FNS dynamics, Lévy superdiffusion, deriving the short-time noise-driven scaling of fractional absolute moments and the monoscaling of self-similar superdiffusion which the model presents. Comparison of each of these dynamics with eye-tracking data is deferred to the next chapter.

2.1 Model setup

Having established the individual spatiotemporal signatures of FNS dynamics, we now define a standard computational framework to test these dynamics against the human behavioral data presented in Chapter 3.

Mathematical formulation of gaze dynamics

The dynamical model of gaze dynamics, inspired by and developed from the FNS theory of neural dynamics, models eye movement trajectories as an underdamped Langevin system driven by an adaptive potential landscape $V_t(\cdot)$ by a kernel of information gathering $k(\cdot)$. Let $\mathbf{x}_t$ denote the two-dimensional position of visual attention, $\mathbf{p}_t$ a generalised auxiliary variable analogous to momentum and encodes short-scale memory effects, $\beta \geq 0$ the coupling strength parameter between $\mathbf{p}_t$ and $\mathbf{x}_t$, $\gamma > 0$ the noise strength parameter as in the standard Langevin system 1.1, and $0 < \alpha \leq 2$ the Lévy tail exponent which defines the heaviness of tails from which noise is drawn, and equivalently the strength of superdiffusion as in Sections 1.2.2 to 1.2.3. The driving noise process dL_t^α represents independent symmetric α-stable ($\mathcal{S}\alpha\mathcal{S}$) Lévy fluctuations.

A deterministic drift $b_\alpha(\mathbf{x}_t)$ drives the system driven by Lévy noise toward an invariant target distribution $\pi(\mathbf{x})$ over long times. As derived in [57], the exact non-local drift for arbitrary $0 < \alpha < 2$ is formulated in terms of the Riesz $(\alpha - 2)$–fractional derivative $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$:

$$b_\alpha(\mathbf{x}) = \frac{\mathcal{D}_{\mathbf{x}}^{\alpha-2}[\pi(\mathbf{x})\nabla_{\mathbf{x}} \ln \pi(\mathbf{x})]}{\pi(\mathbf{x})}. \tag{2.1}$$

where these nonlocal fractional dynamics are treated in detail in Section 2.4. However, due to the computationally intense calculation of the nonlocal operator $\mathcal{D}^{\alpha-2}$, in this work we adopt the local gradient approximation of $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$, which becomes exact as $\alpha \to 2$, as given in Equation (14) of [174]:

$$b_\alpha(\mathbf{x}) \approx c_\alpha \nabla_{\mathbf{x}} \ln \pi(\mathbf{x}), \qquad c_\alpha = \frac{\Gamma(\alpha - 1)}{\Gamma(\frac{\alpha}{2})^2}. \tag{2.2}$$

Combining these components, the full coupled system for gaze dynamics evaluated throughout this

work is given by:

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \nabla V_t(\mathbf{x})dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \nabla V_t(\mathbf{x})dt, \\ \partial_t V_t(\mathbf{x}) &= \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_t), \qquad V_0(\mathbf{x}) = S(\mathbf{x}) \end{aligned} \tag{2.3}$$

where as α, β, γ are parameters modulating FNS dynamics, c_α corrects the scale of the gradient drift, and τ controls information updates to the dynamical-informative potential $V_t(\mathbf{x})$. The nature and purpose of the dynamical updates $k(\cdot)$ is described in Section 2.2. Importantly, due to the equivalent relationship of a Langevin potential in to its targeted distribution in Equation (1.3) and the definition of saliency in visual sampling as in Equation (1.8), we can construct a continuous-time updating internal-state belief distribution of the search target:

$$\pi_t(\mathbf{x}) \propto \exp\left[-V_t(\mathbf{x})\right] \tag{2.4}$$

Empirical protocol from Näher et al.

To validate this modelled computational framework against human behaviour, we emulate the visual foraging task designed by Näher et al. [49], which requires subjects to sequentially scan an array of dots to locate a hidden target. This is done by choosing biologically accurate values for parameters $\alpha, \beta, \gamma, \tau$ and the information gathering scale of $k(\cdot)$. The task itself is encoded in the initial saliency map $V_0(\mathbf{x}) = \log \pi_0(\mathbf{x})$. These parameters and choice of $\pi_0(\mathbf{x})$ are described in the final part of this section.

In the human-testing protocol examined by Näher et al. [49], trials commenced with a pre-search fixation period of duration drawn uniformly between 2500–3500 ms. Following a 100 ms duration of no stimulus, participants were presented with a search array of white dots, each individual dot subtending 0.3° of visual angle on the screen. Exactly 20% of these dots functioned as true targets, which were visually identical to non-targets, and subjects were allotted a maximum search time of 5000 ms to successfully foveate a target, at which point an audio cue would indicate a success, and the subject would be required to fixate for another post-fixation period of 2500–3500 ms. To investigate the influence of spatial structure, the study compared a REGULAR task—where dots were spaced identically to the centers of an invisible hexagonal tiling—with an IRREGULAR task, which randomly redistributed the same number of dots across the screen to break spatial regularity. Assuming a standard viewing distance of 57.3 cm (where 1° ≈ 1 cm) and a monitor pixel density of 36.22 px cm^{-1}, spatial measurements translate directly between pixels and degrees using a scaling factor of 36.22 px $^{\circ -1}$.

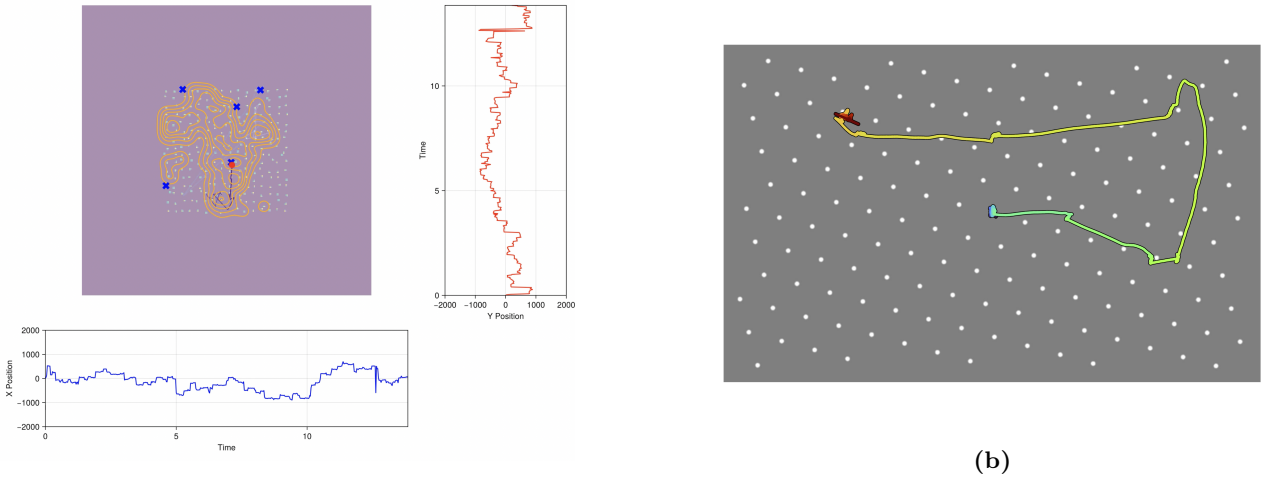
Computational model setup & parameter calibration

The empirical conditions from Näher et al. [49] are translated into numerical model parameters using spatial units of pixels (px) and temporal units of seconds (s). The baseline stimulus potential $S(\mathbf{x})$ is constructed on a 2D grid of Gaussian distributions representing quadratic potential wells over a 16×16 square grid (256 wells total), with 20% randomly designated as targets. Based on fixational eye movement distributions, the standard deviation of each well is set to $\sigma_{\text{well}} = 3$ px. The average axial distance between wells is fixed at 110 px (≈ 3.04° visual angle), bridging the dense (2.3°) and intermediate (4.5°) spacings in the REGULAR human search condition.

The baseline parameters for standard simulation runs are calibrated as follows:

- $\alpha = 1.2$: Captures the characteristic search superdiffusion observed in Section 3.3 while maintaining numerical stability.
- $\sigma_k = 80$ px: Defines an adaptive spatial acuity kernel by the Gaussian $k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right)$. This matches the radial decay scale of human visual acuity (≈ 2.5°, corresponding to 87.2 px at a 50 cm viewing distance with 4 px mm^{-1} density).

- $\tau = 2 \times 10^{-4}$ s^{-1}: Set to yield theta-rhythmic saccadic transitions, as derived in Equation (2.8) and evaluated in Section 3.1. (Note: across simulations, this produces an average ISI of 360 ms, mirroring the empirical saccade rate of 3.53 Hz / 280 ms average ISI).
- $\beta = 1000$ px s^{-1} and $\gamma = 1000$ px^2 s^{-1}: Ensures recovery of gamma-rhythmic harmonic oscillations within the underdamped regime for 3 px wells (Section 3.2).



(a) (b)

Figure 2.1

2.2 Dynamical-information adaptation

The most important component of the dynamics studied here is the *dynamical-informative kernel*. It is the principal extension of fractional neural sampling (FNS) work such as Chen and Gong [11] towards a full computational model of visual search during active vision. We call the kernel dynamical-informative because it couples two processes: the dynamics of the FNS sampler as it explores a landscape, and the information gained during that exploration when it is viewed as an active search. Its development therefore draws on two bodies of work. The first is statistical physics and computational sampling, where an analogous process of kernel deposition is called metadynamics (see Barducci, Bussi, and Parrinello [176] for its introduction and Bussi and Laio [177] for the "well-tempered" variant, which balances sampling accuracy against ergodicity). The second is the psychophysics of how spatial acuity limits information gathering during search.

The section proceeds as follows. Section 2.2.1 states what the kernel must achieve. Section 2.2.2 derives its form from Bayesian information gathering. **??** fixes its spatial scale from the eccentricity dependence of human visual acuity. Section 2.2.3 analyses the resulting dynamics and shows that they encode the theta-rhythmic timescale of saccades studied in Section 3.1, as well as the rate of information gain explored throughout Chapter 4. **??** collects the resulting parameters.

2.2.1 Motivations for kernel adaptation

A dynamic adaptation of the landscape is motivated by three requirements.

1. *Information accounting.* The quantity of information gathered along the gaze trajectory must be tracked, as this is the fundamental purpose of a visual search.

2. *Saccadic regularities.* The model must reproduce the regularities of information-seeking saccades: most importantly the recently discovered strongly theta-rhythmic saccades (Section 3.1), and the well-studied phenomenon of inhibition of return (IoR) [20, 21].

3. *coupling dynamics and searched information.* Any functionally informative description of search dynamics must couple those dynamics to the information gathered so far, completing the feedback loop of active vision [10, 80, 168, 178].

A single mechanism, the continual deposition of a kernel at the gaze position, addresses all three, as the following subsections show.

2.2.2 Kernel adaptation as Bayesian information gathering

In this work the target distribution $\pi_t(\mathbf{x})$ encodes, in a Bayesian sense, the belief that the searched-for target lies at position $\mathbf{x}$ in two-dimensional space. Fixating a location $\mathbf{x}_0$ without finding the target is evidence against the target lying near $\mathbf{x}_0$, so belief must be depleted locally:

$$\pi_{t+\mathrm{d}t}(\mathbf{x}) \propto \pi_t(\mathbf{x})\, e^{-K(\mathbf{x}-\mathbf{x}_0)dt},$$

where $K(\mathbf{x}-\mathbf{x}_0) \geq 0$ is the log-evidence per unit time against discovering a target at $\mathbf{x}$ gathered during $\mathrm{d}t$, which is most informative at $\mathbf{x}_0$ where the most information may be gathered. The dynamics are driven by the effective potential $V_t(\mathbf{x}) = -\ln \pi_t(\mathbf{x})$, so this multiplicative depletion of belief is an additive increment to the potential, $V_{t+dt}(\mathbf{x}) = V_t(\mathbf{x}) + K(\mathbf{x}-\mathbf{x}_0)dt$, up to a spatially uniform constant which normalises the distribution π_t and acts as a scalar gauge symmetry on the potential V_t.

Information gathering is therefore equivalent to raising the potential locally by the information gained at each location. This gives a well-motivated, continuous definition of information gathering, which turns out to be identical to an additive modulation of the potential surface which controls dynamics due to their shared link to an internal belief distribution π_t. The amount gathered at $\mathbf{x}$ depends on its displacement from the gaze position, because acuity falls off away from the fovea (**??**). Writing $K(\mathbf{x})dt = k(\mathbf{x}-\mathbf{x}_0)\,\mathrm{d}t/\tau$ to modulate the timescale use a dimensionless quantity $k(\cdot)$, the instantaneous law of kernel dynamics for gaze at $\mathbf{x}_t$ is

$$\partial_t V_{\mathrm{eff}}(\mathbf{x}) = \frac{1}{\tau}\, k(\mathbf{x}-\mathbf{x}_t), \tag{2.5}$$

where $k(\cdot)$ is an approximately radially symmetric function that tends to zero for large arguments and is scaled so that $k(\mathbf{0}) = 1$. Throughout this work we use an isotropic Gaussian,

$$k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right). \tag{2.6}$$

the two parameters σ_k and τ control the spatial and temporal kernel dynamics respectively. The time constant τ is the time a gaze position needs to raise the potential by one nat at its centre, so $1/\tau$ is the peak rate of information acquisition. The width σ_k is the spatial scale of information gathering. We will see that τ is related to theta-rhythmic activity in gaze dynamics (Section 3.1). We can now show that σ_k is not a free parameter of our model but is set by the acuity of the human eye.

Relation of the kernel width to spatial acuity

The width σ_k is tied to how the detectability of the target falls with retinal eccentricity E, the angular distance from the point of gaze. This is quantified by the visibility map $d'(E)$, measured psychophysically for a given target and background, which ideal-searcher models [10, 74, 97, 168, 179] use to weight the evidence from each fixation in the Bayesian update of target belief [97, 180]. The visibility map declines with eccentricity because due to the decreased resolution outside the central fovea [3–5], through the fall-off of acuity, cortical magnification and crowding, with a half-resolution eccentricity E_2 of a few degrees (about 2.3° in Geisler and Perry [181]) [182, 183]. For an observer with sensitivity d', the expected log-evidence against a target at a target-absent location is $d'^2/2$ per fixation [97, 180], so the kernel is the normalised squared visibility map,

$$k(E) = \left[\frac{d'(E)}{d'(0)}\right]^2,$$

and σ_k is the width of this profile. Thus σ_k is determined by the acuity scaling of the eye and the spatial scale of the target rather than being a free parameter. Calculating the E_2 of our model as the characteristic scale of our Gaussian σ_k, we find a $E_2 = \sigma_k \div 36.22\,\mathrm{px}^{\circ -1} = (80 \div 36.22)^\circ \approx 2.20^\circ$, which is biologically accurate for such a search task as we consider.

2.2.3 Adaptive kernel dynamics as a driver of slow rhythmic exploration

We now predict analytically the rate of excursions under these modelled dynamics in Equation (2.3) due to kernel deposition. The central idea to understand these kernel-driven transitions is to form a tractable transition condition: the effective potential well becomes concave-down at the gaze position. Such a well no longer localises the gaze position $\mathbf{x}_t$, and the Lévy-driven noise will then produce excursions to other modes.

We consider a single isotropic Gaussian mode of the target distribution, centred at zero with standard deviation σ_{fix}, and gaze position fixating within it. In terms of the radial coordinate r about the mode, the effective potential $V_t(r)$ therefore starts as the quadratic well

$$V_0(r) = -\ln \pi_0(\mathbf{x}) = \frac{r^2}{2\sigma_{\text{fix}}^2}.$$

the choice of Gaussian losing very little generality, because a Taylor expansion will show that locally every mode is approximated to second order by a quadratic well—referred to as Laplace's approximation. We assume that the gaze position is localised near $r \approx 0$ and that $\sigma_k \gg \sigma_{\text{fix}}$, so that the small noise-driven motion about the tight potential well with width σ_k can be smoothed over, and kernel deposition can be assumed to be centred at $r = 0$. Excursions are then driven by the deterministic kernel dynamics alone. Lévy noise will introduces Poisson-distributed waiting times between frequent excursions, but analysing these is beyond the scope of this work, and this will only add a slight broadening of the distribution in waiting times modulated by the ratio of noise strength to kernel time constant τ/γ. Under Equation (2.5) with the gaze at the mode centre,

$$\partial_t V_t(r) = \frac{1}{\tau} \exp\left(-\frac{r^2}{2\sigma_k^2}\right),$$

which integrates to

$$V_t(r) = \frac{r^2}{2\sigma_{\text{fix}}^2} + \frac{t}{\tau} \exp\left(-\frac{r^2}{2\sigma_k^2}\right). \tag{2.7}$$

The excursion criterion here is that the radial second spatial derivative of the explored potential at the fixation point $r = 0$ becomes negative, leaving the gaze unconstrained. The second spatial derivative is

$$\frac{\partial^2 V_t(r)}{\partial r^2} = \frac{1}{\sigma_{\text{fix}}^2} + \frac{t}{\tau\,\sigma_k^2}\left(\frac{r^2}{\sigma_k^2} - 1\right) \exp\left\{-\frac{r^2}{2\sigma_k^2}\right\},$$

and at $r = 0$ (where the Hessian is isotropic) it vanishes at the time

$$t_\theta = \tau\,\frac{\sigma_k^2}{\sigma_{\text{fix}}^2}. \tag{2.8}$$

For $t > t_\theta$ the origin is a local maximum of V_t, making this explored region very low interest and unlikely to be explored, which yields a dynamical origin of inhibition of return [20–25]: the gaze is expelled from the location it has just exploited. Strictly, the criterion marks the onset of delocalisation; the excursion to another mode is then carried by the Lévy noise.

It is unsurprising that t_θ is proportional to τ. The dependence on the ratio of variances is of more interest. The dwell time is proportional to the variance σ_k^2 of the adaptive kernel and inversely proportional to the variance σ_{fix}^2 of the mode: a broad kernel takes longer to flatten a given well, and a sharply distributed fixation point will confine the gaze for longer. The result also extends beyond the Gaussian kernel, as σ_k represents the curvature of the kernel function at zero $k(\mathbf{0})$. An extension of

the kernel method to functions which have a cusp at the origin may be biologically realistic [181–183], and could produce a rich description of the drifts and fixational movement of the eyes beyond tremor frequencies [35, 36], but such an approach is less tractable analytically and less stable computationally.

2.3 Local harmonic oscillations

One predictive feature of FNS dynamics is the local oscillations in sampling within confined potentials. We can quantify such high frequency oscillations under FNS dynamics by considering the deterministic part of our FNS equations of motion in Equation (2.3). Considering a Gaussian probability distribution of fixation positions $\pi(\mathbf{x}) \propto \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_{\text{fix}}^2}\right)$ and thus logarithmic gradients $\nabla \ln \pi(\mathbf{x}) = -\frac{\mathbf{x}}{\sigma_{\text{fix}}^2}$. Using this as our saliency landscape $\pi(\mathbf{x})$ without considering perturbation to the explored potential from the slowly updating kernel dynamics, our equations of motion are

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt \end{aligned} \tag{2.9}$$

where we ignore the stochastic noise driving this equation to only consider the coupled ODE which describes deterministic part of $\mathbf{x} = \mathbf{x}_t$ and $\mathbf{p} = \mathbf{p}_t$, where the loss of subscript in typical when moving from an SDE to an ODE

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= -\gamma c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} + \beta \mathbf{p}, \\ \frac{d\mathbf{p}}{dt} &= -\beta c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} \end{aligned} \tag{2.10}$$

where the ODE describing $\mathbf{x}$ can be differentiated, and $\frac{d\mathbf{p}}{dt}$ substituted to give one second order ODE for $\mathbf{x}$

$$\frac{d^2\mathbf{x}}{dt^2} = -\frac{\gamma c_\alpha}{\sigma_{\text{fix}}^2}\frac{d\mathbf{x}}{dt} - \frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2}\mathbf{x} \tag{2.11}$$

Solving the characteristic equation associated with this ODE, $\lambda^2 + \frac{\gamma c_\alpha}{\sigma_{\text{fix}}^2}\lambda + \frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2} = 0$, we have

$$\lambda = \frac{1}{2}\left(-\frac{\gamma c_\alpha}{\sigma_{\text{fix}}^2} \pm \sqrt{\frac{\gamma^2 c_\alpha^2}{\sigma_{\text{fix}}^4} - 4\frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2}}\right) \tag{2.12}$$

Assuming an underdamped regime where the discriminant is negative, the resonant frequency ω of the fixational dynamics corresponds to the' imaginary part of the eigenvalues

$$\omega = \sqrt{\frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2} - \frac{\gamma^2 c_\alpha^2}{4\sigma_{\text{fix}}^4}} \tag{2.13}$$

which yields ordinary frequency

$$f = \frac{1}{2\pi\sigma_{\text{fix}}}\sqrt{\beta^2 c_\alpha - \frac{\gamma^2 c_\alpha^2}{4\sigma_{\text{fix}}^2}} \tag{2.14}$$

2.4 Lévy superdiffusion

To model the superdiffusion of a computational exploratory dynamical system such as found in searching eye movements, our system is driven by Lévy noise withing an over-damped Langevin regime (see Sections 1.2.1 to 1.2.3). Such underlying dynamics are known as Lévy flight (see 1.2.3) and are the paradigm superdiffusive system. However, both the potential-driven adaptive kernel dynamics, and auxiliary momentum dynamics add complexity to the superdiffusive properties of the model. Quantifying superdiffusion in this system exploring a confined and dynamic potential landscape requires

a dedicated analysis of how Lévy dynamics will introduce monoscaling superdiffusion which is self-similar on all scales, with the same superdiffusive patterns driving large jumps across a landscape as those small-scale jumps when considering finer movements. Lévy motion unifies these scales under a single superdiffusive pattern, motivating our unification of gaze dynamics under a single model.

Analysis of Lévy flight superdiffusion from the fractional Fokker-Planck equation

Lévy processes are the canonical form of monoscaling superdiffusion, yet the analysis of the precise form of diffusive scaling requires close analysis. Lévy flight superdiffusion is analysed from the deterministic dynamics of the probability distribution of jump size $P(\|\Delta\mathbf{x}\|)$ dynamics of the probability distribution. This problem of understanding the dynamics of the distribution of $\mathbf{x}$ is identical to that approached when studying classical Brownian motion where the classical Fokker-Planck equation was developed (see 1.2.1). Here in the case of a generalised Lévy-driven Langevin system with momentum coupling, the deterministic dynamics of the distribution of $\mathbf{x}$ is governed by the fractional Fokker Planck equation, as described in [57, 184, 185]. We can consider the phase space of a generalised coupled set of n stochastic differential equations of d-dimensions of $\mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_n]^T$, where $d\mathbf{z}_t = (\mathbf{D}+\mathbf{Q})\, b(\mathbf{z}_t)dt + \mathbf{D}^{\frac{1}{\alpha}} d\mathbf{L}_t^\alpha$ for $(n\times d)$–dimensional Lévy noise. Here, $\mathbf{D}$ is the positive semi-definite matrix which defines strength and coupling of noise across $\mathbf{z}$, and $\mathbf{Q}$ represents the skew-symmetric matrix defining dynamical coupling across $\mathbf{z}$. For such a system the fractional Fokker-Planck equation can be expressed exactly:

$$\partial_t P(\mathbf{z},t) = -(\mathbf{D}+\mathbf{Q})\nabla_{\mathbf{z}}\left[b(\mathbf{z})P(\mathbf{z},t)\right] - \mathbf{D}\mathcal{D}^\alpha_{\mathbf{z}} P(\mathbf{z},t) \tag{2.15}$$

where $\mathcal{D}^\alpha_{\mathbf{z}}$ represents the α-fractional Riesz derivative, which generalises the operation of the Laplacian to fractional orders, and is defined in Fourier space as $\mathcal{D}^\alpha_{\mathbf{z}}[f] = \mathcal{F}^{-1}\|\mathbf{k}\|^\alpha \mathcal{F}[f]$, with the case of $\alpha = 2$ restoring the ordinary Fokker-Planck equation with $\mathcal{D}^{\alpha=2}_{\mathbf{z}} = -\nabla^2_{\mathbf{z}}$. It can be shown that our model of gaze dynamics is recovered for $\mathbf{z} = [\mathbf{x}, \mathbf{p}]$, $\mathbf{D} = \begin{bmatrix}\gamma & 0\\ 0 & 0\end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix}0 & -\beta\\ \beta & 0\end{bmatrix}$, shown in [57]. The $b(\mathbf{z})$ drift term required to recover a given distribution is presented in Equations (2.1) to (2.2). Any solutions to the dynamics in Equation (2.15) are nontrivial, as the fractional derivative results in a nonlinear multivariate integro-differential equation which is difficult to propagate, even numerically.

When considering diffusion over short timescales, the physics of fractional Langevin systems with $\alpha > 1$ are noise dominated. In our Lévy driven system, short times are dominated by α-stable dL^α_t Lévy noise, across power-law distributed spatial scales of short jumps and nonlocal jumps to distant positions in short times. This is corroborated by the scaling of Lévy noise as $dL^\alpha_{\Delta t} = \Delta t^{1/\alpha} dL^\alpha_1$ in Equation (1.6) is stronger than ballistic effects which scale in short times as Δt. Thus, considering the fractional Fokker-Planck equation evolving from a delta function initial condition $P(\mathbf{z},0) = \delta^n(\mathbf{x}-\mathbf{x}_0)\delta^n(\mathbf{p}-\mathbf{p}_0)$ evolving through a quadratic potential using the position coordinate $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}_0$, yielding a drift term $\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}(\mathbf{x}_0) + \mathcal{O}(\Delta\mathbf{x})$. We can construct a leading-order short time solution up to $t = \Delta t$ and marginalise over the auxiliary momentum $\mathbf{p}$ which in practise is a nuisance variable, we have:

$$\partial_t P(\Delta\mathbf{x},t) = -\gamma\mathcal{D}^\alpha_{\mathbf{x}} P(\Delta\mathbf{x},\Delta t) - \nabla_{\mathbf{x}}\cdot\left[\left(\gamma\mathbf{b}(\Delta\mathbf{x}) + \beta\,\mathbb{E}[\mathbf{p}\,|\,\Delta\mathbf{x},\Delta t]\right)P(\Delta\mathbf{x},\Delta t)\right] \tag{2.16}$$

where we can truncate these terms to highest order using

$$\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}_0 + \mathcal{O}(\Delta\mathbf{x}), \qquad \mathbb{E}[\mathbf{p}|\Delta\mathbf{x},\Delta t] = \mathbf{p}_0 + \mathcal{O}(\Delta t), \qquad \mathbf{v}_0 = \gamma\,\mathbf{b}_0 + \beta\,\mathbf{p}_0 \tag{2.17}$$

which gives

$$\begin{aligned}
\partial_t P(\Delta\mathbf{x},\Delta t) &= -\gamma\mathcal{D}^\alpha_{\mathbf{x}} P - \nabla_{\mathbf{x}}\cdot\left[\left(\mathbf{v}_0 + \mathcal{O}(\Delta\mathbf{x}) + \mathcal{O}(\Delta t)\right)P\right]\\
\partial_t P(\Delta\mathbf{x},\Delta t) &= -\gamma\mathcal{D}^\alpha_{\mathbf{x}} P - \mathbf{v}_0\cdot\nabla_{\mathbf{x}} P + \left[\mathcal{O}(\Delta t^0) + \mathcal{O}(\Delta t^{1-1/\alpha})\right]\cdot P\\
\partial_t \hat{P}(\mathbf{k},\Delta t) &= -\gamma\|\mathbf{k}\|^\alpha \hat{P} - i\,\mathbf{v}_0\cdot\mathbf{k}\,\hat{P} + \mathcal{O}(\Delta t^0) \qquad (2.18)\\
\implies \hat{P}(\mathbf{k},\Delta t) &= \exp\left[-\gamma\|\mathbf{k}\|^\alpha\Delta t - i\mathbf{v}_0\cdot\mathbf{k}\,\Delta t\right]\left[1+\mathcal{O}(\Delta t)\right] \qquad (2.19)
\end{aligned}$$

Showing that, to first order in Δt, the short-time diffusion of under our model Equation (2.3) yields a $\mathcal{S}\alpha\mathcal{S}$ distribution which is instantaneously transported with velocity $\mathbf{v}_0 = \gamma\mathbf{b}_0 + \beta\mathbf{p}_0$.

Quantification of Lévy superdiffusion monoscaling

Following on from the solution for the characteristic equation $\hat{P}(\mathbf{k}, \Delta t) = \exp\left[-\gamma\|\mathbf{k}\|^{\alpha}\Delta t - i\mathbf{v}_0 \cdot \mathbf{k}\,\Delta t\right][1+\mathcal{O}(\Delta t)]$, we reduce to the characteristic function of Fourier-space jump sized $k = \|\mathbf{k}\|$. For $0 < q < 2$ this change of variables will give the relation [186]

$$\langle\|\Delta\mathbf{x}\|^q\rangle = C_q \int_0^{\infty} \left[1 - \operatorname{Re}\hat{P}(k, \Delta t)\right] k^{-1-q}\,\mathrm{d}k, \qquad C_q = \frac{q(1-q)}{\Gamma(2-q)\cos(q\pi/2)}, \tag{2.20}$$

where C_q is defined at $q = 1$ by its limit, $C_1 = 2/\pi$. Substituting our result of Equation (2.19) into Equation (2.20) and evaluating after a further change of variable $u = \gamma\,\Delta t\,k^{\alpha}$, we have for $0 < q < \alpha$:

$$\langle\|\Delta\mathbf{x}\|^q\rangle = C_q \left[\frac{\Gamma\left(1-\frac{q}{\alpha}\right)}{q}(\gamma\Delta t)^{q/\alpha} + \frac{\Gamma\left(\frac{2-q}{\alpha}\right)}{2\alpha}\frac{v_0^2}{\gamma^2}(\gamma\Delta t)^{(2-\frac{2}{\alpha})+\frac{q}{\alpha}} + \mathcal{O}\left(\Delta t^{1+q/\alpha}\right)\right]. \tag{2.21}$$

where the first term is the moment of an α-stable Lévy flight with scale $(\gamma\Delta t)^{1/\alpha}$, which is finite only for $q < \alpha$ [187]. The second term is the ballistic contribution of the momentum, which will tend to dominate as oscillations over longer times. Since $2 - \frac{2-q}{\alpha} > \frac{q}{\alpha}$ for $\alpha > 1$, it is subleading at short times, effecting dynamics more as $q \to 0$ and $\alpha \to 1$. This gives diffusion through fractional absolute moments scaling as

$$\langle\|\Delta\mathbf{x}\|^q\rangle \propto \Delta t^{\lambda(q)}, \qquad \lambda(q) = \frac{q}{\alpha}, \qquad 0 < q < \alpha. \tag{2.22}$$

Thus, for $q < \alpha$ the modelled dynamics of Equation (2.3) reproduce the scaling of a Lévy flight [175]. Such superdiffusion will produce a self-similar process with a single scaling exponent $1/\alpha > 1/2$ across all scales. This is called monoscaling superdiffusion, where superdiffusion is present and identical across all scales of q.

However, for $q \geq \alpha$ the moments of an unbounded Lévy flight diverge and the expansion above no longer applies. It is thus important to computationally examine the behaviour of a Lévy flight during exploration of particular potentials, whether or not they are unbounded or dynamic, and if any post-processing is done to handle divergences in the variance of jumps. In other studies, Lévy flights have been found to have $\lambda(q) \neq \frac{q}{\alpha}$, and are thus proposed to be multifractal with different scaling behaviour for large jumps as for small. This is the signature of strong anomalous diffusion [188], however it is misleading, as a Lévy flight is defined by monoscaling self-similarity. In this study, we find that under the confined dynamics and light low pass filtering to produce continuous trajectories, the Lévy flight yields a single monoscaling superdiffusive exponent of $\lambda(q) = \frac{q}{\alpha}$

Chapter 3: Empirical Validations of Model Predictions

Our model is designed to be compared against empirical data and against established accounts of the phenomenon it describes, rich gaze dynamics during visual search. In this chapter, we compare each of the modelled dynamics of Chapter 2 with high precision eye-tracking data gathered by collaborators in Näher et al. [49] as described in Section 2.1. gaze dynamics during visual search show many broad and often task-dependent spatiotemporal patterns, so we ask which are most important for understanding computation in the brain. The literature inspires our partition of modelled dynamics, namely 2.2 information rhythmicity, 2.3 local short-time oscillatory sampling, and 2.4 global spatial superdiffusion, maps onto important visual dynamical signatures that have been observed elsewhere. These correspond to (1) theta-rhythmic macrosaccades, (2) gamma-scale 40–100 Hz fixational ocular drift, and (3) global saccadic superdiffusion to an informative visual scene [83, 189]. Importantly, our unifying model provides an understand of each of these dynamics as a function of network-level neural activity, including the novel connection between microtremor and the neural gamma signal, theta-gamma coupling, and the monoscaling of Lévy superdiffusion across spatial scales.

Section 3.1 examines theta rhythmicity in saccades, comparing the model's transitional rhythmicity with the saccade timing in the data. Section 3.2 examines gamma-scale oscillations during visual fixations, testing the model's local high-frequency harmonic oscillations against fixational drift. Section 3.3 examines superdiffusion in search, assessing whether the global spatial statistics of gaze are consistent with Lévy superdiffusion. Finally, Section 3.4 presents the spatiotemporal predictions of the model across these scales and discusses how they unify the experimental and modelling findings above.

3.1 Theta rhythmicity of saccades

The approximate 200 ms timescale of human saccades has long existed experimentally [50, 190–192]. A direct measure of saccade rhythmicity is the direct measurement of the spread of the ISI distribution, assuming a large lognormal spread of intervals Δt. This can be used to construct a measure called the Rhythmicity Index (RI):

$$\mathrm{RI} = \frac{1}{\mathrm{CV}_{\ln}}, \qquad \mathrm{CV}_{\ln} = \sqrt{e^{\sigma_{\ln}^2} - 1} \tag{3.1}$$

where $\sigma_{\ln}$ is the standard deviation of the distribution of ISIs in log-space $\ln \Delta t$, with $\mathrm{CV}_{\ln}$ often called the geometric CV. This RI is an effective measure of the spread of a strictly positive distribution. A Poisson distributed variable will give $\mathrm{RI} = 1$, whereas it is found that for the ISI distribution of subjects considered here for a random NOISE condition is $\mathrm{RI} = 2.69$ [49].

3.1.1 Local rhythmicity of saccades

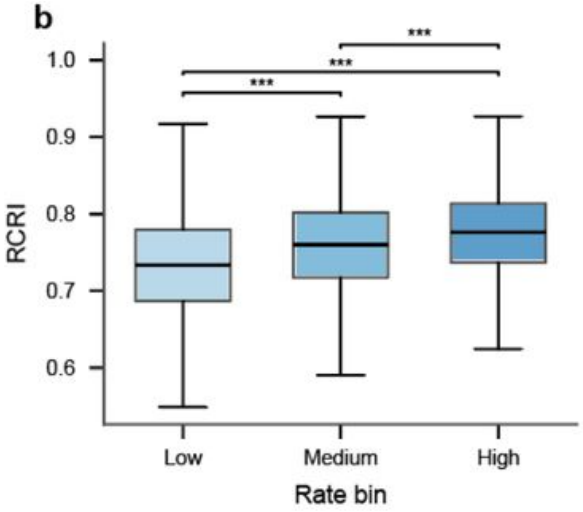
The recent discovery of the strength of saccade rhythmicity by Näher et al. [49] and others [29] required a deep understanding of the rhythmicity of biological systems. A challenge that is often present in biological systems is that timescales are not driven by a perfect clock, but instead the characteristic timescales of certain regular patterns in the brain. As such, the 200 ms estimate for saccades during a visual search is not expected to be a universal timescale across subjects and trials. Instead, the model used for such processes in neuroscientific works is a noisy rise-to-threshold model, where a random walk is constructed, and when it reaches a goal, a decision is made [49] [CITE]. The noise-driven rise for this process is expected to have a rate which is slowly varying about a characteristic timescale, in the case of saccades around 2–6 Hz. Since this slowly varying rate is expected to be stable for a single individual and a single trial, we consider a local measure of rhythmicity for the intervals between the saccades within a timseries, called the Inter-Saccadic-Intervals (ISI). Such a metric can

be proposed—as developed by [49]—with the local Rate-Corrected Rhythmicity Index (RCRI):

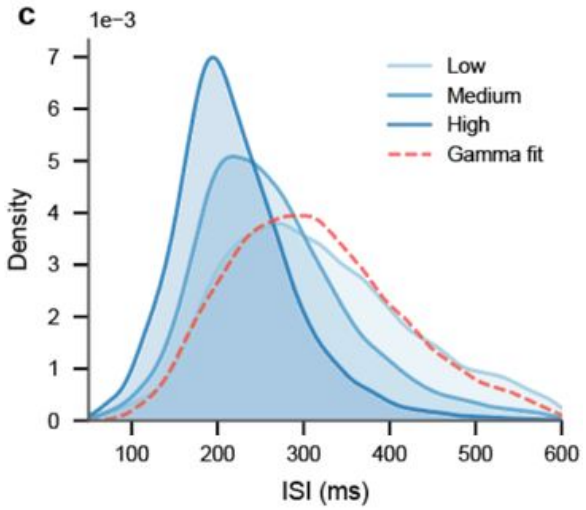
$$\mathrm{RCRI} = \frac{1}{1 + \langle \mathrm{CV}_2 \rangle}, \qquad \mathrm{CV}_2(i) = 2\frac{\|\Delta t_{i+1} - \Delta t_i\|}{\Delta t_{i+1} + \Delta t_i} \tag{3.2}$$

which allows quantification of purely local rhythmicity, comparing each interval Δt_{i+1} to it's predecessor Δt_i, and averaging across all intervals considered i, usually a single subject. The results of such an analysis have resulted in two key conclusions. Firstly, human saccades are very strongly rhythmic, with RCRI ≈ 0.76 for the subjects as described in Section 2.1 depending on condition. Secondly, it has been found that the instantaneous saccade rate is a strong predictor for the local saccade rhythmicity RCRI, with a correlation calculated at $\rho = 0.39, P = 0.001$, with saccade rate found to be a significant predictor of RCRI with $\beta = 0.0114\,\mathrm{Hz}^{-1}, P = 0.014$

those particular individuals and trials binned with a higher saccade rate correlate with higher RCRI and thus stronger comparable rhythmicity [49], as shown in Figure 3.1a.



(a) Rate-corrected rhythmicity index (RCRI) is positively correlated with saccade rate. Collaborators Näher et al. [49] uncovered this statistically significant result, which implies that shorter inter-saccadic intervals (ISIs) will result in more regular ISIs, which corresponds to stronger kernel dynamics as defined in Section 2.2.3.



(b) The probability distribution of inter-saccadic intervals (ISIs) are tightly distributed in a Gamma-10 distribution. Our collaborators in Näher et al. [49] find that saccades are more locally rhythmic than previously considered. In particular, the ISI distribution is a tight Gamma-10 distribution, but the ISI timeseries is strongly locally rhythmic, as captured by the rate-corrected rhythmicity index (RCRI).

Figure 3.1

Interpretations of local rhythmicity and saccade rate for a model of gaze dynamics

These results support our unified stochastic-informative formulation of gaze dynamics. We have shown good agreement between the tight distribution of modelled ISIs, more closely locally. We have also introduced a natural parameter τ with which to modulate the slowly varying theta-scale saccade rate. This also connects the dynamics of information gathering with the dynamics of regular attention in a search task. The results that rhythmicity is strongest where a task contains nearly identical searched objects is encouraging for this information-driven saccade rate theory. A correlation between high saccade rate and high rhythmicity is predicted from the origin of regular dynamical-informational transitions as described in Section 2.2, where a stronger kernel produces more rhythmic transitions, and a weaker kernel will approach an arrhythmic Poisson point distribution of ISIs, reducing the RCRI, as predicted.

3.2 Gamma oscillations during visual fixation

The second most important time- and spatial scale of gaze dynamics are those of high frequency fixational dynamics. Typically fixational dynamics are divided into microsaccades, low-frequency linear drift, and high frequency ocular tremor. There is evidence that microsaccades are driven by the same statistics and process as regular (called macro-) saccades, and linear drifts have been found to be important in gathering information. An account of low-frequency linear drift easily fits into the model presented here, for instance as steep peak in the dynamical kernel or a simple drift in the potential landscape, producing a self-avoiding walk as described in [7] and described in Table A.1.

The current study is the first to make a well-motivated prediction concerning the neural origin and structure of high-frequency ocular microtremor. Identifying ocular tremor as high frequency gamma-scale visual sampling is a novel and useful approach in understanding perceptual sampling. Ocular tremor, under this hypothesis, will originate in the neural control of visual information sampling, likely in the brainstem where eye movements originate [CITE].

3.2.1 Previous treatment as ocular microtremor

The high frequency ocular tremor is the smallest fixational motion, only existing with amplitude of around one minute of arc. Although, it is nonetheless recently shown to be very important in dynamically sampling high contrast details and high spatial frequencies in images. and is increasingly studied as measurement equipment are non-invasive and designed to accurately and measure low-amplitude high-frequency ocular signals.

3.2.2 Analysis of fixational Fourier spectra

This is the first study to examine the high frequency fixational tremor during visual search. Other studies have constructed artificial tasks where subjects are instructed to fixate on a point of interest for extended periods, often stabilising the head with bite-bars. Such setups are not designed to represent natural gaze dynamics, especially during a search task. The computational difficulty in extracting such low-amplitude, high frequency signals from the data in [49] is twofold: Firstly, the fixations only contain an average of around 200 ms of data, and leaving only an approximate 10 periods of such high frequency signals, and secondly such high frequency signals are often error-prone when examining noise-containing positional data.

We are lucky in that we have a large amount of data with a very high quality, sampled at a temporal sampling frequency of 1000 Hz, and with arcsecond precision [CITE PROPER VALUE]. Since the data will then contain low frequency drifts, we expect a power spectral density to scale as $\mathrm{PSD}(\mathbf{x}) \sim f^{-2}$, and thus when analysing, we correct this slope by applying a finite-differencing time derivative to extract velocity, where the null hypothesis of no significant tremor peak would result in white noise $\mathrm{PSD}(\mathbf{v}) \sim 1$.

Our data pipeline here is thus to apply a light 4th-order Buterworth low-pass filter at 200 Hz to handle high-frequency instrumental noise, extract 2D velocity for each eye through central-differencing the positional data, and performing Fourier analysis on such velocity data.

3.3 Search superdiffusion

Superdiffusion is often a prominent pattern in natural dynamical systems, where its existence implies some self-similar fractal structure. Such dynamics have not been the focus of other studies, and often visual trajectories have been considered as a fixational Brownian motion externally interrupted with saccades. We here present novel multiscale superdiffusive pattern in the diffusion of eye movements from eye trajectory data. Many studies which have described eye movement dynamics as normally diffusive stems from the sensitivity of data processing choices in finding anomalous diffusion exploring confined boundaries such as fixations. Conventional analyses often inadvertently obscure critical spatiotemporal features.

The most important methodological choice in accurately characterising visual diffusion patterns is the choice to time-average displacements, and choosing a short timescale over which diffusion scaling is considered. As demonstrated by Vilk et al. [133], aggregating data across disparate temporal regimes can wash out localised scaling behaviours, yielding an artefact of normal diffusion. Studies which construct regimes of normal and subdiffusive regimes are thus common when examining eye movements, but this ignores the continuous-time superdiffusion which we unify under multiple scales. For such studies, see [39, 72, 193, 194]

Analysis of multiscale superdiffusion in eye-tracking data

We analyse the binocular-averaged gaze trajectories over all 22 human subjects provided from the study in Näher et al. [49], as described in Section 2.1. From the onset of the search task, we time-average the fractional absolute displacements $\langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle$ for 30 values of q linearly separated from $q = 0.1$ to $q = 3.0$. These fractional absolute displacements are evaluated across all valid time lags $\Delta t < 1000\,\text{ms}$ with a resolution of $\Delta t = 1\,\text{ms}$. All intervals containing NaN values (such as blinks) are ignored. We then ensemble-average these displacements across all subjects.

We perform a linear best-fit in log-log space of these absolute fractional displacements to find power-law diffusive scaling with time, thus sampling from our moment spectrum function $\lambda(q)$ as defined in Equation (1.7). Each of these fractional absolute displacements are shown in **??**. The moment spectrum function $\lambda(q)$ thus is found to have a good fit to $\lambda(q) = (0.893 \pm 0.006)q$ with an error in the fit of $R^2 = 0.999$. This behaviour gives monoscaling across the shortest to longest spatial scales of fixations to saccades. The corresponding Lévy exponent given such linear monoscaling in $\lambda(q)$ is thus found to be $\alpha = \frac{q}{\lambda(q)} = 1.120 \pm 0.007$.

3.3.1 Spectrum of fractional moments

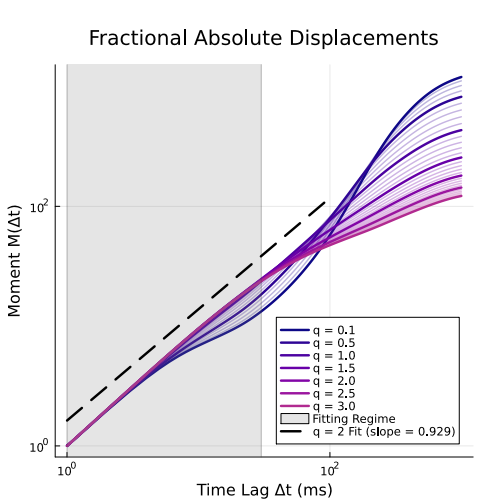
Figure 3.2 summarises the scaling of the time-averaged fractional absolute displacements. For comparison of the various scaling exponents, we define a rescaled moment function $M_q(\Delta t)$ to give good comparison of the spectrum of moments. $M_q(\Delta t)$ is here defined as

$$M_q(\Delta t) = M_q(\Delta t_{\min})^{-1} \langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle^{1/q} \sim \Delta t^{\lambda(q)/q} \tag{3.3}$$

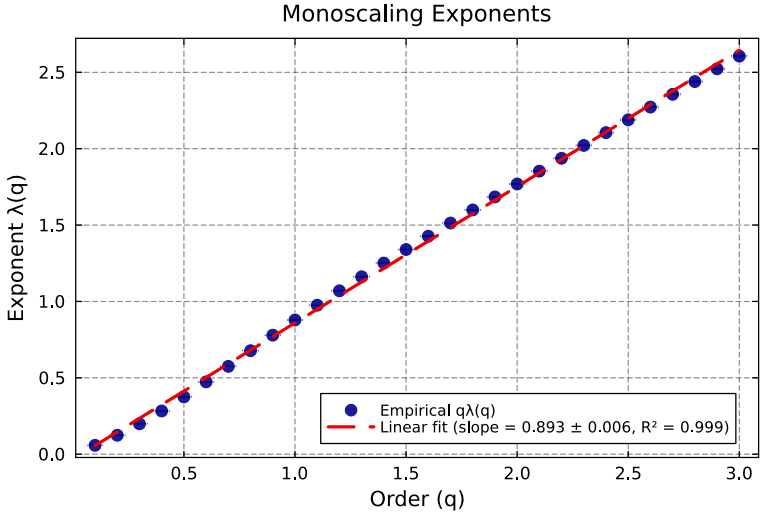
with C_q defined such that $M_q(\Delta t_{\min}) = 1$ to give a shared origin at the first time lag Δt. Small q gives strong weight to the small, frequent displacements of fixational motion, while large q weights the rare, large displacements which are given in saccades. The shape of $\lambda(q)$ therefore tests whether one exponent may describe superdiffusion on all scales. A linear spectrum, $\lambda(q) = q/\alpha$, is monoscaling: all moments grow with the same exponent, as for a self-similar Lévy process with stability index α. A nonlinear spectrum is the general case found in many systems called strong anomalous diffusion which consist of two dynamical strategies stitched together at different scales [133, 188, 195–197]

Figure 3.2a shows the rescaled moments $M_q(\Delta t)$ for the human eye-tracking data. Under this rescaling, the slope of each curve in log–log space is $\lambda(q)/q$. Monoscaling therefore predicts that these curves share a single slope across the whole spectrum of q, with $\frac{\lambda(q)}{q} = 1/\alpha$. For comparison, Brownian motion ($\alpha = 2$) has slope $1/2$ and ballistic motion has slope 1 across the entire spectrum q. The monoscaling exponent is found by performing a linear regression to our spectrum $\lambda(q)$. Figure 3.2b shows these fitted exponents $\lambda(q)$ against q. The linear fit is good, with $\lambda(q) = (0.893 \pm 0.006)\,q$ and

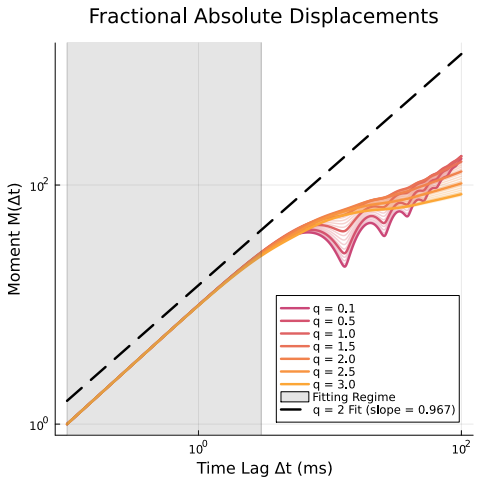
$R^2 = 0.999$, across the whole considered range $0.1 \leq q \leq 3.0$. This gives a Lévy index $\alpha = q/\lambda(q) = 1.120 \pm 0.008$, which is well below the Brownian value of 2 and so indicates strong superdiffusion. Because the spectrum is well described by a linear regression, there is no crossover in scaling between the low-q moments that probe fixational motion and the high-q moments that probe saccades. A single exponent describes both, validating the unifying model derived from FNS dynamics.



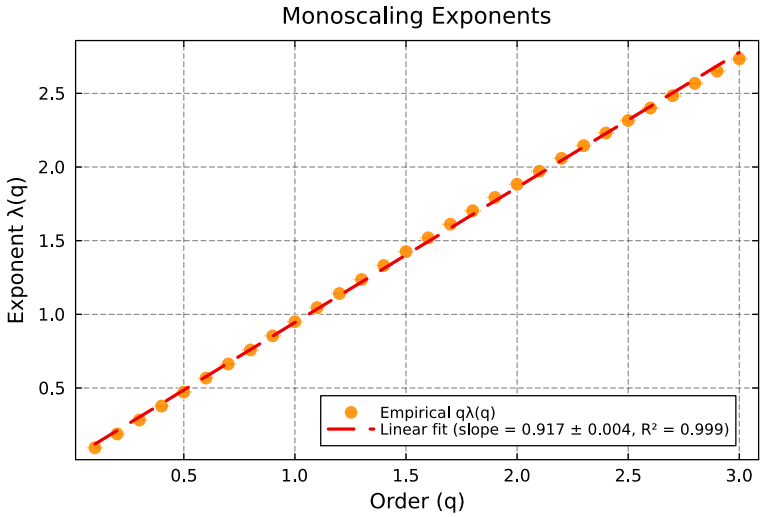
(a) Data: rescaled moments



(b) Data: spectrum $\lambda(q)$



(c) Model: rescaled moments



(d) Model: spectrum $\lambda(q)$

Figure 3.2: Monoscaling superdiffusion in human visual search trajectories and in the model. Time-averaged fractional absolute displacements $\langle \|\Delta\mathbf{x}\|^q \rangle$, which generalise the mean squared displacement to a spectrum of orders q, with small q emphasising small fixational displacements and large q emphasising saccades. The top row shows binocular-averaged gaze data, with moments ensemble-averaged over all 22 subjects of Näher et al. [49] (approximately 10,000 s of data; exact sample size to be updated). The bottom row shows the same analysis applied to trajectories simulated from the model of Chapter 2. **(a,c)** Moments rescaled as $M_q(\Delta t) \propto \langle \|\Delta\mathbf{x}\|^q \rangle^{1/q}$ and normalised to $M_q(1\text{ ms}) = 1$, for 30 values of q from 0.1 to 3.0. In log–log space the slope of each curve is $\lambda(q)/q$. **(b,d)** Scaling exponents $\lambda(q)$ from power-law fits to the curves on the left, up to $\Delta t = 30$ ms. For the data, the relation is linear, $\lambda(q) = (0.893 \pm 0.006)\, q$ with $R^2 = 0.999$, so all moments share a single scaling exponent. This corresponds to a Lévy stability index $\alpha = q/\lambda(q) = 1.120 \pm 0.007$, well below the Brownian value $\alpha = 2$, indicating strong superdiffusion from fixational scales to saccades. For the model, the fit gives $\lambda(q) = (\,0.917 \pm 0.004\,)\, q$, i.e. $\alpha = 1.091 \pm 0.005$.

Implications of monoscaling superdiffusion

The most important implication of this particular superdiffusion results is that the strong monoscaling of visual search trajectories with Lévy index $\alpha = 1.120$ is robustly multiscale. The robust linear scaling of $\lambda(q) = \frac{q}{\alpha} = \frac{q}{1.120}$ across a spectrum of q-values $0 < q < 3$ is important, as it presents an identical process driving both fixational eye movements — including tremor, drifts, and microsaccades — with small relative values of $\|\Delta\mathbf{x}\|$ emphasised with low values of q, as well as large nonlocal saccades with large relative values of $\|\Delta\mathbf{x}\|$, emphasised through large values of q. Importantly, this validates our construction of a continuous-time unified model of eye movements, unified across all spatiotemporal scales of visual search.

3.4 Spatiotemporal predictions from the model

The above have been analyses of data which already exist in the literature to varying degrees of novelty. In particular, the approximate timescale of saccades has been known for the history of eye movement research, whereas the significance of the rate-corrected local rhythmicity is a very recent result only discovered in the year of this present work, 2026. Ocular microtremor as a phenomenon has been found in many studies, however consensus has never been reached as to it's importance, and it has never been linked to a computational role of gamma-scale rhythms. Lastly, the close mathematical analysis of search superdiffusion has only been recently undertaken, and where it has been studied it has been done naively as a simple Lévy walk with no awareness of the non-equilibrium potential landscape being explored. However, another benefit of comparing the generative unifying model presented in Chapter 2 to the data is that further dynamical predictions outside of the present literature can be made. Thus, we present experimental paradigms and features of this model which are being developed to better understand visual searching.

3.4.1 Saliency-frequency coupling in contrast-dependant search tasks

The ocular microtremor is here hypothesised to be harmonic oscillations in the control of the path of the eye during fixation tasks. An experiment may be designed in which targets with various degrees of contrast are presented to be searched. High contrast, localised points, such as tight, high spatial frequency Gabor patches, would produce localised saliency wells. Whereas, low contrast, large low spatial frequency patches would produce saliency wells with a large width σ_{fix}. High contrast wells would correspond to shorter theta timescales, quadratic in the effective saliency well size as in Equation (2.8), and a microtremor rhythm which is approximately inverse to the effective saliency well size as in Equation (2.13). It could this be examined if these dynamics are coupled, by experimentally saving trials with three data points: t_θ, f_{peak}, and k_{Gabor}. A simple regression analysis can be achieved here to find correlations between spatial frequency k_{Gabor} and the visual theta-rhythm t_θ and the f_{peak}. It can then also be tested if these effects are consistent with a potential well of variable size, such that $t_\theta \propto \sigma_{\text{fix}}^{-2}$ and $f \propto \sigma_{\text{fix}}^{-1}$

3.4.2 Smooth pursuit predictions and parametric-fitting techniques

Another experimental paradigm which provides insight into the behaviour of the control of visual trajectories. In particular, the exact same model can be examined in the behavioural paradigm of smooth tracking of a moving object. In particular, a typical experiment would be the tracking of a smoothly moving pendulum. It is realistically found that a typical eye movement trace contains discontinuous micro-saccades during smooth pursuit, and discontinuous catch-up saccades. In particular, we expect from the model that the instability will be most present in the periods of stability when kernel deposition is strongest. Further experiments could also find the behaviour of dynamics in phase space $\alpha/\beta/\gamma/t_\theta$ for different neurological conditions. For instance, it is well-known that schizophrenia effects the

Chapter 4: Functional Advantages of Visual Search Dynamics

The strength of a well-motivated biologically realistic model of eye movements is in examining functional roles of individual dynamical components studied in the rich task of gathering information. This approach allows an understanding of brain function to be inferred from the dynamics of visual search, as the map between neural sampling dynamics and visual searching dynamics is explicit through this work. A major motivation for this chapter is the previous work in studying the optimal computation of the visual system and broadly of the brain; firstly, that saccade locations are well predicted by greedy information-maximisation strategies [168, 198, 199], and secondly that this efficiency and function of gaze dynamics during search is a significant indicator of brain health and function [62–65, 67, 68, 93]. This forms a novel connection between the abstract performance of probabilistic sampling and direct metrics of the dynamics of visual search which act as a proxy for network-level brain function.

The first section of this chapter (Section 4.1) presents informational-theoretic methods which have been constructed to analyse efficiency during abstract search tasks. Key to this method is the quantification of information of a belief distribution as the distribution entropy $S_{\pi_t} = -\int_{\mathbf{x}} d\mathbf{x} \ln \pi_t(\mathbf{x})$, and tracking its development through a search task. We take these metrics and compare the informational advantages of each dynamical pattern by comparing an abstract sampler operating in different points in parameter space $\alpha/\beta/\gamma/\tau$. The final section (**??**) examines further extensions of these dynamics beyond the case of 2-dimensional visual search to high-dimensional or dynamical target landscapes, where control and exploration are hypothesised be more responsive and robust to shifts in a loss function.

4.1 Functional roles of dynamical patterns

The dynamics studied in this thesis are characterised by three distinct spatiotemporal signatures: gamma-band oscillations, theta-scale rhythmicity, and superdiffusive spatial trajectories. The framework of Chapter 2, constructs a biologically-motivated model to produce these signatures through an abstract search task. In Chapter 3 we found that the same dynamics have been found to be important in studying real human gaze trajectories, especially during realistic search tasks. In this section we make use of the theoretical foundations of this model as a sampling theory, and identify the functional advantages of each pattern in an abstract sampling process which the brain would be trained to complete: local exploitation in gamma oscillations, temporal organisation in theta rhythmicity, and global exploration through Lévy α-superdiffusion. To make these roles precise, we define a set of information-theoretic metrics for search efficiency and, for each metric in turn, describe how each of the three patterns effects it.

The three patterns are controlled by distinct parts of the model:

- **Gamma oscillations** arise from the memory–momentum term p_t (strength β) and the α-Lévy noise dL_t^α within a highly localised potential well. They are analogous to the gamma oscillations found throughout the brain in almost every computational task, and to high-frequency ocular microtremor in gaze.

- **Theta rhythmicity** is the regular, theta-scale timing of transitions between potential wells, tied to the rate of information acquisition through the informative timescale τ (Equation (2.8)). For motion driven by Lévy or Brownian noise alone, transition intervals are instead exponentially distributed according to a Poisson point process, as follows from the Lévy–Itô decomposition theorem (Section 1.2.3).

- **Superdiffusion** is governed by the stability index α and sets the spatial scaling of global exploration. Superdiffusive paths are thought to be optimal for exploring sparse unknown landscapes under the Lévy foraging hypothesis (Section 1.2.2).

4.1.1 Information gain

When considering the efficiency advantages of strategic sampling in a visual search task, the central quantity under study is the information gain, $IG(t)$. To quantify informational search efficiency across an abstract search landscape, we take inspiration from Bayesian inference and sampling methods. In particular, the adaptive dynamics of our sampling model closely resemble well-tempered metadynamics in Markov chain Monte Carlo (MCMC) sampling [200]. Similar techniques have been used to model visual search in discrete time by Bujia et al. [17], where the equivalent target distribution is the likelihood distribution for a search goal, and information is tracked only at the goal position. In Section 2.2 we expanded and unified these approaches by analysing a continuous-time dynamical-information kernel. By tracking the posterior target distribution after kernel deposition at every location in the visual scene, we model global information acquisition across an evolving landscape of belief.

Given a search through a probabilistic landscape for high-probability modes, the information extracted is proportional to the change in entropy of the target distribution. Since we study updates to an effective belief distribution, we use the language and tools of Bayesian inference directly. The initial pre-search belief distribution, $\pi_0(\mathbf{x})$, is defined through the saliency-potential landscape $V_0(\mathbf{x})$ by Boltzmann–Gibbs statistics,

$$\pi_0(\mathbf{x}) = \frac{\exp\{-V_0(\mathbf{x})\}}{\int_{\mathbf{x}'} \exp\{-V_0(\mathbf{x}')\}\, d\mathbf{x}'}. \tag{4.1}$$

As the search progresses, the potential updates in continuous time to encode learned information, through an accumulated kernel over time t,

$$V_t(\mathbf{x}) = V_0(\mathbf{x}) + \int_0^t k(\mathbf{x}_{t'})\, \frac{dt'}{\tau}. \tag{4.2}$$

This adapted potential defines the posterior belief distribution, the updated target likelihood landscape at time t,

$$\pi_t(\mathbf{x}) = \frac{\exp\{-V_t(\mathbf{x})\}}{\int_{\mathbf{x}'} \exp\{-V_t(\mathbf{x}')\}\, d\mathbf{x}'}. \tag{4.3}$$

A natural first step in comparing information gathering through time is the cross-entropy of the posterior to the prior,

$$H(\pi_t, \pi_0) = -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x}. \tag{4.4}$$

However, cross-entropy alone is of limited utility, as it conflates the internal concentration of belief within the posterior (its self-entropy, $H(\pi_t)$) with the net information gained relative to the prior. To decouple these contributions we subtract the posterior entropy to obtain the Kullback–Leibler (KL) divergence $D_{\mathrm{KL}}(\pi_t \parallel \pi_0)$, which gives the formal definition of information gain up to time t:

$$\begin{aligned} IG(t) &= H(\pi_t, \pi_0) - H(\pi_t) \\ &= -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x} + \int \pi_t(\mathbf{x}) \log \pi_t(\mathbf{x})\, d\mathbf{x} \\ &= \int \pi_t(\mathbf{x}) \log \left(\frac{\pi_t(\mathbf{x})}{\pi_0(\mathbf{x})} \right) d\mathbf{x} \\ &\equiv D_{\mathrm{KL}}(\pi_t \parallel \pi_0). \end{aligned} \tag{4.5}$$

The KL divergence is non-negative and asymmetric. In the context of an abstract sampling model, non-negativity reflects the fact that any spatial update from π_0 represents accumulated evidence, though not necessarily optimal search efficiency. A weak kernel, a slow informative timescale τ, or deposition in low-likelihood areas alters the landscape only marginally and results in slower growth in IG. The information content of the search trajectory is therefore entirely captured by IG, which

quantifies the global information acquired as the observer's belief landscape evolves from π_0 to π_t. The metrics in the following subsections summarise different aspects of the curve $IG(t)$.

Since τ sets the timescale of information gathering, time-dependent metrics such as IV and FPT are trivially improved by decreasing τ. When testing the effect of a dynamical pattern, particularly theta-rhythmic transitions, comparisons must therefore be made at fixed τ or through dimensionless metrics.

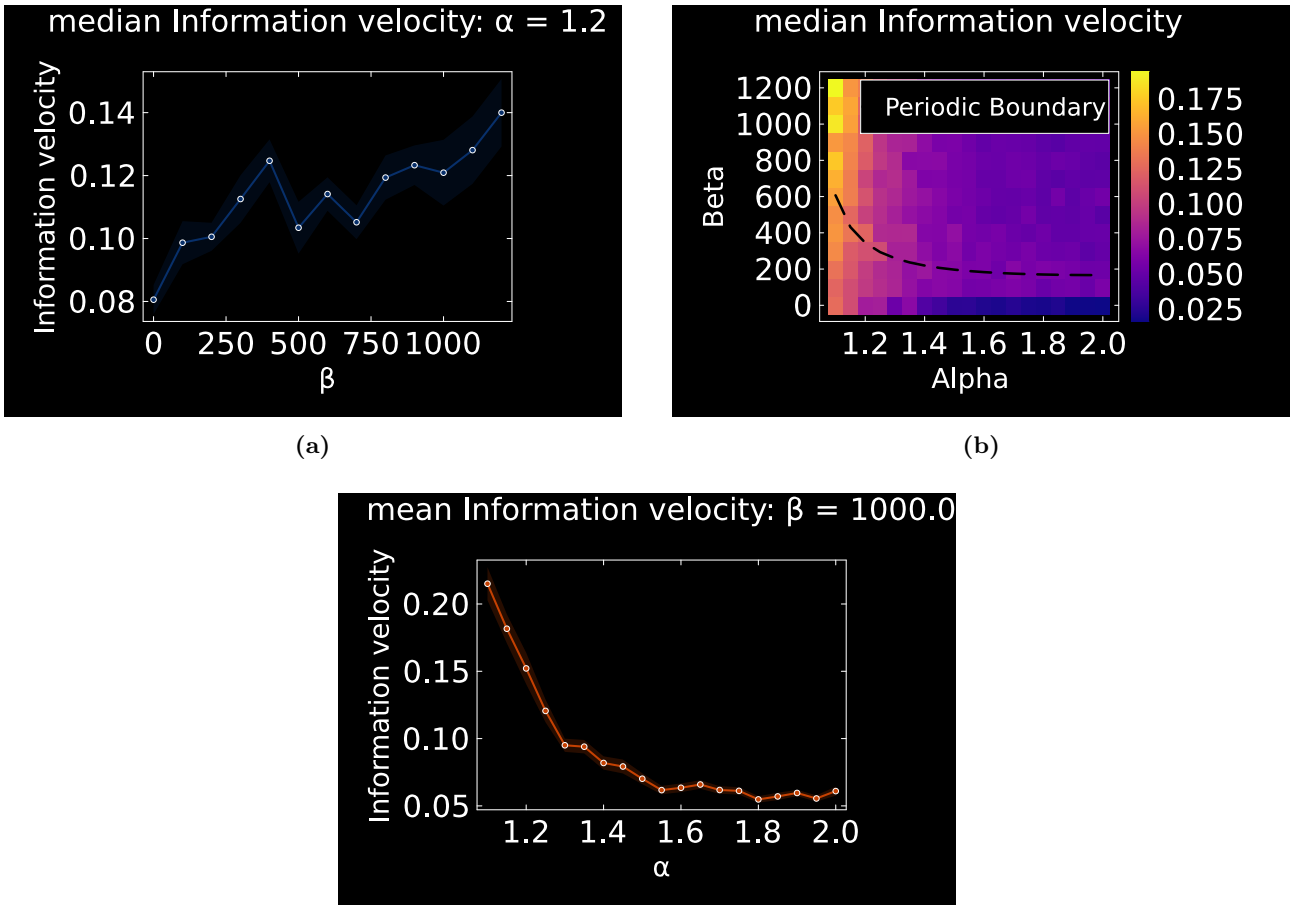
4.1.2 Information velocity

The information velocity, $IV = dIG/dt$, is the first derivative of the information gain. Comparing its averaged value at the same informative timescale τ measures the quality of information gathered by different strategies. A strategy that targets high-likelihood areas, and drives transitions sensitive to decreases in the level of likelihood, will have high average IV_{mean} and IV_{med}.

Gamma oscillations. Rapid local decorrelation means kernel is deposited across the well at many independent locations, so we expect stronger oscillations (larger β) to increase how quickly information is extracted from a high-likelihood region (Figure 4.1a,b).

Theta rhythmicity. Rhythmic transitions end each dwell at a point linked to information acquisition, so time is not wasted in depleted wells. At fixed τ this is expected to raise the average information velocity relative to Poisson-distributed transitions.

Superdiffusion. Long jumps place the walker in new high-likelihood regions where kernel deposition is most informative, which is expected to raise the mean IV relative to purely local diffusion (Figure 4.1c).



(a) **(b)**

(c)

Figure 4.1: Information velocity. (a) Median mean IV against the momentum strength β at $\alpha = 1.2$. (b) Heatmap of median mean IV. (c) Mean IV against the stability index α at $\beta = 1000$.

4.1.3 First passage time

Perhaps the clearest metric of search efficiency is the speed of completing a task. In our system this is the duration of each trial, the first passage time (FPT). We can derive an expected FPT from the theta-scale rate of saccades (Equation (2.8) in Section 2.2) and the number of target points in the visual scene relative to the total. For K targets among N points, the probability of reaching no target after n saccades is the hypergeometric distribution evaluated at zero,

$$p_n(0) = \frac{\binom{K}{0}\binom{N-K}{n}}{\binom{N}{n}} = \frac{(N-K)!\,(N-n)!}{(N-K-n)!\,N!}. \tag{4.6}$$

The expected number of saccades $\mathbb{E}[n]$ is the n for which Equation (4.6) first falls below 0.5, and the expected first passage time is $\mathbb{E}[\mathrm{FPT}] = t_\theta\,\mathbb{E}[n]$. For our standard parameters and experiment, defined in Section 2.1, this gives

$$\begin{aligned}\mathbb{E}[\mathrm{FPT}] &= t_\theta \times \mathbb{E}_{N=256,K=5}[n] \\ &= 0.1422 \times 94 \\ &= 13.36 \text{ s}.\end{aligned} \tag{4.7}$$

This calculation assumes that each transition is independent and drawn from a stationary distribution with mean t_θ across the search task. The true dynamics of the system are therefore expected to yield an averaged FPT that depends on the system parameters, and Equation (4.7) serves as a baseline against which the effect of each pattern is assessed.

Gamma oscillations. The memory–momentum term β keeps the sampler responsive to real-time changes in the saliency landscape as information is received from the scene. This responsiveness is not captured by the baseline, and can shorten the time to find a target.

Theta rhythmicity. The baseline uses the theta rate t_θ directly, so theta rhythmicity sets the mean waiting time per saccade. Regular intervals with the same mean are expected to narrow the spread of first passage times compared with exponentially distributed intervals.

Superdiffusion. A superdiffusive walker has access to the entire scene, and may explore nearby fixation points before migrating to far-displaced areas. This breaks the independence assumed in the baseline, so the mean FPT is expected to depend on α through how efficiently the walker reaches unvisited targets (Figure 4.2).

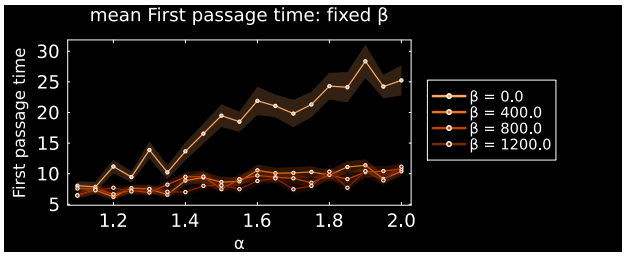


Figure 4.2: First passage time. Mean FPT against the stability index α.

4.1.4 Shape parameter

While the preceding metrics depend explicitly on the search timescale, we can evaluate the fundamental efficiency of a sampling process by making it dimensionless. By normalising each trial so that both time and information content range from 0 to 1, we can examine the shape of the monotonic accumulation curve $IG(t)$ independently of absolute time. Statistically, this is captured by the area under the normalised curve, which can be transformed into an effective power-law exponent for the growth of information gain.

Gamma oscillations. Gamma activity acts at short times, so its influence on the dimensionless curve is expected to appear in how front-loaded information gain is within each visited well.

Theta rhythmicity. Because the shape parameter is dimensionless, it separates the effect of rhythmic transitions from the trivial dependence on τ. Regular transitions avoid long dwells in depleted wells, which should keep the normalised accumulation curve closer to linear rather than saturating early.

Superdiffusion. Trapping in a depleted well causes information gain to plateau early, giving a strongly concave normalised curve. By escaping such wells, superdiffusion is expected to sustain information gain later in the trial and so raise the effective exponent (Figure 4.3).

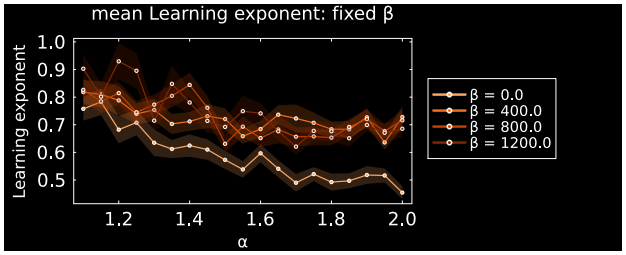


Figure 4.3: Shape parameter. Mean shape exponent of the normalised information-gain curve against the stability index α.

4.1.5 Gini coefficient

To measure the inequality in the information gained, we bin the trial into time windows longer than the gamma oscillation and compute the Gini coefficient of the information gained across bins. A low value indicates that information is acquired evenly through the search, while a high value indicates that it is concentrated in a few periods.

Gamma oscillations. Since the bins are longer than the gamma period, the oscillations themselves are not resolved. The metric instead tests whether stronger local sampling spreads information acquisition more evenly across successive windows (Figure 4.4).

Theta rhythmicity. This is the metric most directly tied to theta organisation. Because rhythmic transitions end each dwell at a point linked to information acquisition, rather than at a random time, we expect a more even distribution of information across time windows, and hence a lower Gini coefficient than for the Poisson case.

Superdiffusion. Long jumps may concentrate information in bursts when a new mode is reached, which would raise the Gini coefficient for small α.

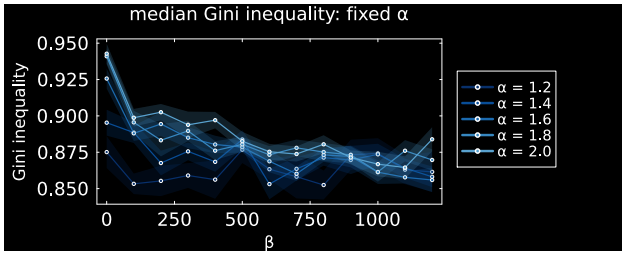


Figure 4.4: Gini coefficient. Median Gini coefficient of binned information gain against the momentum strength β.

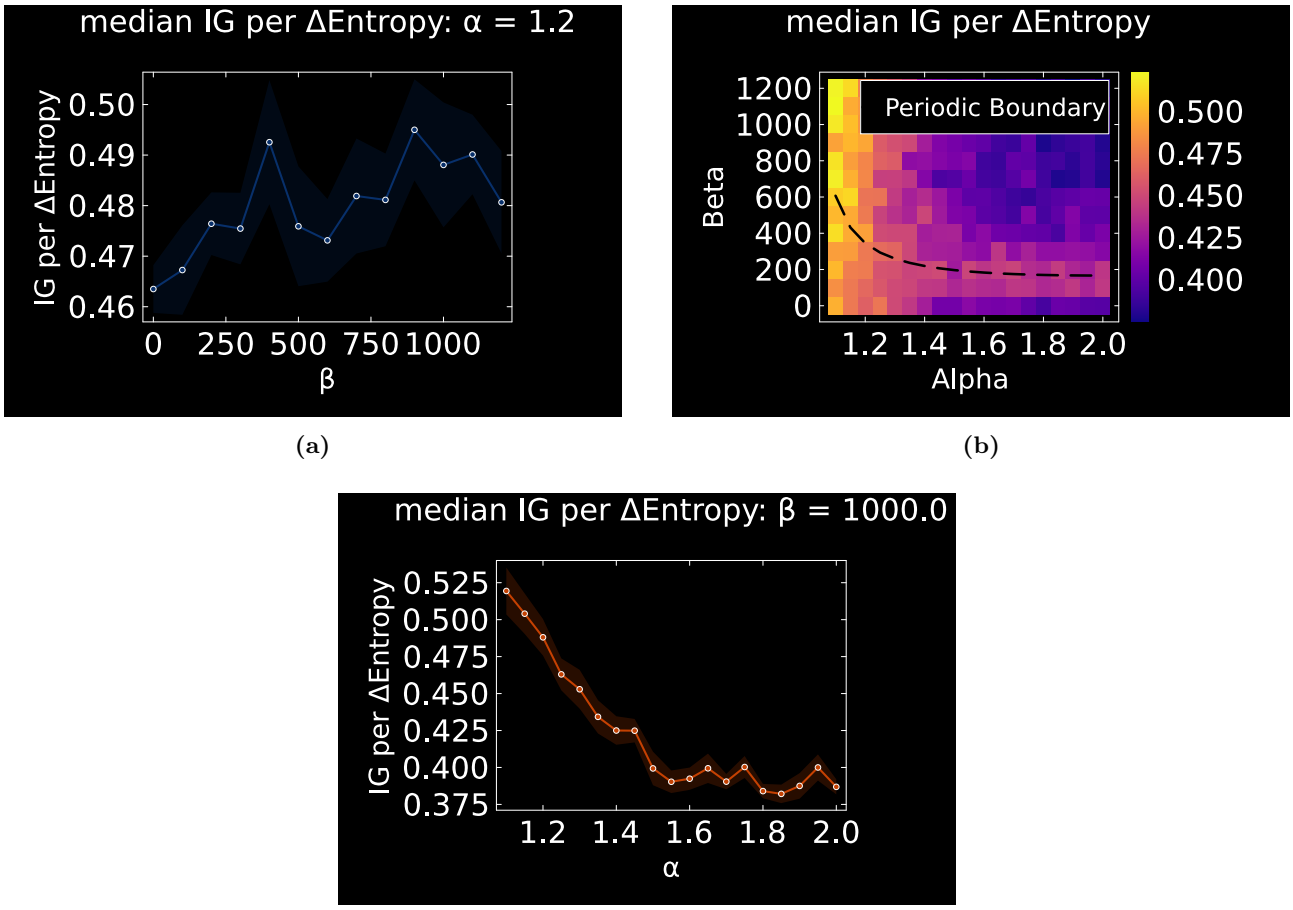
4.1.6 Information–entropy gradient

The information–entropy gradient, $\mathrm{IG_H}$, measures the gradient of information gain with respect to changes in posterior certainty, that is, the efficiency with which a change in belief gains information.

Gamma oscillations. Sampling local gradients directs the deposition of the kernel towards where the posterior is changing, so we expect each unit of belief change to yield more information, raising IG_H (Figure 4.5a,b).

Theta rhythmicity. Theta rhythmicity governs when sampling moves rather than where, so its effect on IG_H is expected to be indirect, through how fully each well is exploited before transition.

Superdiffusion. By visiting unexplored modes, superdiffusion reduces posterior entropy where it is least certain, so we expect it to change the information gained per unit of entropy (Figure 4.5c).



(a) **(b)**

(c)

Figure 4.5: Information–entropy gradient. (a) Median IG_H against the momentum strength β at $\alpha = 1.2$. (b) Heatmap of median IG_H. (c) Median IG_H against the stability index α at $\beta = 1000$.

4.1.7 Summary

Table 4.1 collects the expected effect of each pattern on each metric. Gamma oscillations act at the finest scale to probe local gradients, theta rhythmicity organises when sampling transitions between wells, and superdiffusion sets where the sampler can go. Together they cover complementary aspects of the information-gain curve.

Metric	**Gamma oscillations**	**Theta rhythmicity**	**Superdiffusion**
IV	Faster local information extraction	Avoids wasted time in depleted wells	Accesses new high-likelihood regions
FPT	Responds to landscape changes	Sets mean waiting time; narrows spread	Enables global access and escape from local traps
Shape	Front-loads information gain within wells	Keeps accumulation closer to linear at fixed τ	Sustains information gain later in the trial
Gini	Tests evenness across time windows	Promotes more evenly distributed information gain	May produce bursts of information at mode changes
IG_H	Supports gradient-directed deposition	Acts indirectly through well exploitation	Changes information gain through exploration of new modes

Table 4.1: Expected effect of each dynamical pattern on each metric. Entries summarise the proposed mechanisms and are to be confirmed against the results.

Chapter 5: Discussion

In this thesis we have presented the first model to unify the rich multiscale dynamics of visual search. Previous work has focused their study to models of either large-scale saccades or small-scale fixations. Further, motivations for such studies have been divided in either aiming to present a biologically accurate understanding of visual search, or a full mechanistic generative model of eye-tracking data. Taking aim to unify these motivations and spatiotemporal scales of study, we constructed a model from the neural dynamics developed under the theory of fractional neural sampling (FNS) [11, 57]. Under this framework, gaze dynamics are governed by a second-order stochastic process (Equation (2.3)) driven by Lévy noise exploring a dynamical saliency landscape adapts through a dynamical-informative kernel (Section 2.2). We understand the model dynamics through three key spatiotemporal signatures: global α–Lévy superdiffusion, rhythmic transitions between potential wells driven by an adaptation from a kernel function $k(\cdot)$, and local high-frequency harmonic oscillations driven by a memory–momentum parameter β. Each of these signatures maps onto a well-known behaviour during gaze dynamics. We compared each of these dynamics with the high-resolution gaze data of Näher et al. [49] and with the wider literature, and found that the model captures the signature spatiotemporal dynamics of saccades and fixations within a single framework (Chapter 3). In particular, we discovered robust monoscaling Lévy superdiffusion on all spatial scales with $\alpha = 1.12$, theta-rhythmic saccades with $t_\theta \approx 200\,\text{ms}$ and a high degree of rhythmicity RCRI $= 0.72$, and local high-frequency harmonic oscillations with mean frequency $f = 56.5\,\text{Hz}$. Finally, after validating the biologically-motivated model of gaze dynamics, we formulated functional explanations for each of their existence. Using information-theoretic metrics on the updating belief distribution present in the model, we show that each pattern plays a distinct functional role in the efficiency of informative search (Chapter 4).

The methods and dynamics here studied are of great importance to physicists as a description of a non-equilibrium complex dynamical system. First, visual dynamics are shown to be superdiffusive, and moreover display robust monoscaling, governed by a single stability index α. Such anomalous diffusion at every scale is of interest when studying how basic computational dynamics may be constructed from complex dynamical systems at criticality. We characterised the short-time, noise-driven scaling of such biological data through fractional absolute moments (Section 3.3), and compared these to the monoscaling that an implementation of Lévy dynamics can provide (Section 2.4). Such rich multiscale self-similarity has a physical explanation as the result of a complex dynamical network at criticality with diverging correlation length, resulting in nonlocal spatial access to an informative landscape. Second, the model presents and explains the unique pattern of nested oscillations: theta-scale rhythmic transitions between regions, which modulate gamma-scale sampling dynamics, arising from neural adaptation, and the confinement of a sampling process. Nested timescales are a recurrent feature of coherent neural dynamics, especially during communication of information between areas of the brain [9, 28, 103–106]. The present model carries this framework over to gaze dynamics, as they emerge from the same neural dynamics which control the process of active vision. Third, the dynamical-informative kernel introduced in this work describes how visual dynamics are influenced by what is perceived, so the searcher depletes information as it samples an informative distribution π in search for a target. Our formulation naturally encodes this information through a belief landscape which evolves from a prior saliency π_0 to a posterior belief after exploration π_t. The cumulative information gathered is then well defined through the Kullback–Leibler (KL–) divergence. Such a well-motivated physics-inspired development of interpretable memory-effects, heavy-tailed noise, and nonlocality is important in the development and progress of physics as an interdisciplinary field.

This work advances the study of the brain in neuroscience, as our model turns gaze into a source of interpretable parameters modulating neural attention, a very important topic of study when understanding decision-making in the brain. The active parameter space here defined through the stability index α, memory–momentum strength β, noise scale γ, and informative timescale t_θ; each parameter has a well-understood and motivated have a defined effect on the trajectory, so fitting them to data estimates the dynamics of the underlying attentional control. Several experiments would sharpen

this. High-frequency eye tracking without a bite-bar, combined with head acceleration measurements, would resolve gamma-scale fixational dynamics in naturalistic, head-free viewing and separate eye from head contributions. High-quality smooth pursuit experiments, in which the target moves, would probe how the landscape is tracked over time. Dynamical search tasks with controlled likelihood landscapes would constrain the kernel and the theta-rhythmic transitions. Because gaze aggregates the activity of many regions, the fitted parameters may summarise collective properties of the circuits that control and process visual input. A natural candidate is the interaction between the superior colliculus, which integrates sensory, cortical, and goal-related signals into a map that drives saccades [[**CITE**]], and the visual cortex, which supplies the evidence on which that map is built [[**CITE**]]. Within the active vision loop, the model's evolving potential could be compared with such a priority map, and the kernel with processes such as inhibition of return [20, 21], giving a way to study attention in this system quantitatively. Because alterations of eye movements accompany many neurological and psychiatric conditions [[**CITE**]], mechanistic parameters of this kind also offer a route to more accurate biomarkers, and potentially to treatments that target a specific part of the loop.

This work advances the study of the brain in neuroscience, as our model turns gaze into a source of interpretable parameters modulating neural attention, a very important topic of study when understanding decision-making in the brain. The active parameter space here defined through the stability index α, memory–momentum strength β, noise scale γ, and informative timescale t_θ; each parameter has a well-understood and motivated effect on the dynamics and function of visual sampling, so fitting them to data estimates the dynamics of the underlying attentional control of the aggregated network of the brain.

Several experiments may be developed to apply this framework to understand brain health in particular. An experimental protocol tracking high-frequency eye movement alongside coarse-grained saccade positions during a standardised search task without head restraint, combined with head acceleration measurements, would resolve gamma-scale fixational dynamics in naturalistic, head-free viewing and separate eye from head contributions. During such an experiment, coherence of the high-frequency sampling during fixations can be examined rigorously, as well as testing the saliency-frequency coupling and mapping the paramter-space dynamics as described in Section 3.4. Such an experiment is simple to design using new low-cost and non-invasive techniques such as described in [34] and the data extracted is tractable to analyse across the scales of interest here, to test the predictions of FNS dynamics here presented.

High-quality smooth pursuit experiments, tracking a dynamical target, would probe dynamical attention. The present model is easily extended to a dynamical potential, where information adaptation is modelled to forget recent information as it becomes less reliable in a dynamical environment. As such, a potential may be represented as:

$$V_t(\mathbf{x}) = V_{\text{dynamic}}(\mathbf{x}; t) + K(\mathbf{x}; t), \qquad \partial_t K(\mathbf{x}; t) = \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_t) - \frac{1}{\tau'} K(\mathbf{x}; t) \tag{5.1}$$

where such dynamics will constrain the noise produced, for instance a slow-moving pendulum is predicted to have high rate of large micro-saccades at the apex, whereas a faster pendulum will slow the rate of kernel deposition, making microsaccades as likely as during the smooth pursuit of the pendulum. Because gaze aggregates the activity of many regions, the fitted parameters during any given task will provide an aggregate understanding of the neural circuits that control and process visual input, which are a significant part of the brain. A natural candidate for locating these neural interactions is between the superior colliculus, which integrates sensory, cortical, and goal-related signals into a map that drives saccades [[**CITE**]], and the visual cortex, which supplies the evidence on which that map is built [[**CITE**]]. Because dynamics of eye movements accompany many neurological and psychiatric conditions [[**CITE**]], mechanistic parameter fitting of this kind also offer a route to highly accurate, simple, and non-invasive biomarkers.

The same sampling process applies to machine learning, in particular to sampling accurately from a given posterior distribution $\pi_{\text{post}}(\boldsymbol{\theta})$ over a set of parameters $\boldsymbol{\theta}$ using a set of methods called Markov chain Monte Carlo (MCMC). Standard MCMC methods are based on sampling a chain of points in

parameter space, and proposing local moves through a probability landscape $V(\boldsymbol{\theta}) = -\ln \pi_{\text{post}}(\boldsymbol{\theta})$, often incorporating identical or derived stochastic Langevin dynamics as introduced in Section 1.2. Such dynamics will struggle to robustly explore and mix between well-separated modes, and designing an algorithm to robustly explore a wide range of heavy-tailed or multimodal distributions efficiently remains a long-standing and active topic in classical machine learning [172, 173, 206]. The model's superdiffusive jumps address this directly, and its adaptive kernel resembles well-tempered metadynamics [200], in which deposited bias discourages revisiting explored regions, robustly improving ergodicity without reliance on parameter tuning on the unknown landscape $V(\boldsymbol{\theta})$. The link to free-energy minimisation and active inference [**[CITE]**, 70] is here important, since both treat search as the reduction of uncertainty about a landscape.

Our results emphasise that coupling oscillations across scales is itself important for an abstract sampler: rapid local oscillations explore each mode, while slower rhythmic transitions decide when to move on. Superdiffusive dynamics in generalised computational systems have recently attracted considerable interest, both from theoretical studies of such dynamics emerging from informational sampling [135, 207–209] as well as in designing algorithms in MCMC sampling of abstract distribution landscapes [173, 174, 185, 204, 205]. The same principle of heavy-tailed nonlocality in computation has also influenced more general AI frameworks, notably in work from our group on Fractional Neural Attention (FNA) [210]. FNA is designed to token interactions through Lévy diffusion, as governed by the fractional Laplacian, so that short- and long-range dependencies arise from a single operator rather than hand-designed local–global patterns or fine-tuned scales of influence. In the reported experiments it matches or exceeds Transformer baselines on text classification, image classification, and machine translation, in some cases with a single layer, and its attention networks show larger spectral gaps and shorter path lengths, consistent with faster information mixing. That the same mechanism describes both gaze and attention in artificial networks supports the view that multiscale, nonlocal sampling is a general computational strategy.

In conclusion, this thesis has shown that the rich and apparently disparate dynamics of human gaze can be described by a single physical model with a small set of interpretable components which derive from neural dynamics: nonlocal Lévy jumps, nested oscillations, and an adaptive, information-gathering search landscape. Gaze is both easily measured and serves as a strong proxy for neural decision-making and attention, so gaining an understanding of gaze dynamics gives access to basic computation in the brain. The model offers a quantitative framework making use of concepts from the statistical physics of diffusive processes, makes testable predictions for new experiments to test the theory, and provides a process through which we can understand neural attention at a large scale, and interpret it to develop and understand brain health and basic function of abstract dynamical computational systems.

Appendix A: Prior models of visual search

Model	Core domain	Primary aim	Key features
Ideal Bayesian searcher (IBS) Najemnik and Geisler [97], see also [74, 179]	Information- and decision theory	Establish an optimally informative process for saccade localisation based on biologically realistic information foraging of the eye.	Exact Bayesian belief updating, incorporating biologically realistic retinal resolution gradients, spatial noise, and global entropy reduction. Considers discrete saccades rather than continuous time.
Least action / active inference Zanca and Gori [70]	Machine learning	Generating scanpaths (trajectories) through physics-inspired variational informational free-energy minimisation.	Generative deterministic model of scanpaths. Sophisticated account of natural image saliency. Does not consider fixational dynamics, does not aim to be biologically realistic.
Self-avoiding random walk Engbert et al. [7]	Statistical physics and oculomotor control	Develop an accurate account of low-frequency ocular drifts during free-viewing fixations.	Lattice-based stochastic motion with self-avoidance constraints; 2D quadratic potential. Does not capture a high-frequency tremor or saccades.
DeepGaze Abdelrahman, Al-Tawil, and Al-Hamadi [85], see also [18, 75, 88, 89]	Deep learning and computer vision	Construct a neural network to predict human scanpaths on natural images with high empirical accuracy.	Modular deep neural network trained on eye-tracking data; models internal state, memory, and information. Not designed as a biologically realistic model; considers discrete saccade locations.
Itti–Koch–Niebur model Itti, Koch, and Niebur [94]	Biologically inspired vision	Simulate bottom-up, stimulus-driven visual attention based on early primate visual pathways.	Multi-scale feature extraction (colour, intensity, orientation); winner-take-all selection with inhibition of return (IOR).
Feature-integration theory Gayet and Aydemir [96]	Cognitive psychology	Explain experimental differences in performance during sparse- and dense-search tasks.	Pre-attentive direction of attention based on saliency theory. Biologically motivated, but not generative or explanatory for eye movement measurement.
Guided search Wolfe [95] and Wolfe, Cave, and Franzel [211]	Cognitive psychology	Account for the continuum of multi-scale search efficiency by combining bottom-up and top-down guidance of attention.	Parallel feature maps weighted by stimulus salience and target template to form an activation map; serial deployment of attention in decreasing activation order.

Table A.1: Review of a selection of influential models of visual search and gaze dynamics. Comparison across core theoretical domains, primary objectives, and defining operational features. Models span a spectrum from biologically grounded frameworks prioritising physiological mechanisms (e.g., Feature-Integration Theory, Itti–Koch–Niebur) to normative and computational architectures optimised for empirical trajectory prediction on complex natural scenes (e.g., Ideal Bayesian Searcher, DeepGaze). Important features which share analogy in the model presented in this thesis include: an information kernel (in discrete time) from the approach in Najemnik and Geisler [97], the second-order (noiseless) dynamical system coupled to a rich account of saliency gradients presented in Zanca and Gori [70], and the description of the rich non-tremor fixational random motion in the presented in Engbert et al. [7].

Appendix B: Code availability

Repository	Function
`VisualFNS.jl`	Producing simulations, visualisations, and raw data of gaze trajectories through numerically integrating the system Equation (2.3) with kernel adaptation under Equation (2.5). Important techniques in solving this system were the use of the Euler–Maruyama numerical scheme for SDE integration and the Fourier-space spectral solution of kernel updates.
`VisualFNS_Analysis.jl`	Analysing trajectory data which was output from `VisualFNS.jl`. Involved producing power spectra, marking intervals between saccade-like transitions, and analysing superdiffusive scaling.
`VisualData_Analysis.jl`	Loading in the data provided from Näher et al. [49], and analysing it in the same way as in `VisualFNS_Analysis.jl`. Involved producing power spectra, marking intervals between saccade-like transitions, and analysing superdiffusive scaling. Originally forked from `FractionalSaccades.jl`.
`FractionalNeuralSampling.jl`	The project which provides basic methods to formalise the FNS equation and its variants in the syntax solvable through `DifferentialEquations.jl`.

Table B.1: A list of repositories containing the code used in this thesis. All code was written in Julia, apart from some helper scripts written in Python.

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